Testing SWINGERsim

Part II: Three published multiple stars – BU CMi, TIC 120362137 and V994 Her

2026-11-06

Contents

1 Aim of the test **2**

2 Data and methods **3**
2.1 The three systems . . . 3
2.2 Terms used in this report . . . 4
2.3 Three model configurations . . . 4
2.4 The model . . . 5
2.5 Comparing light curves . . . 5
2.6 The fit . . . 5
2.7 Ensembles . . . 6

3 BU CMi (2+2) **6**
3.1 The system according to Pribulla et al. (2023) . . . 6
3.2 Stability . . . 6
3.3 A century forward and backward, and the apsidal motion . . . 7
3.4 TESS light curves . . . 9
3.5 Times of minima . . . 11
3.6 The fit . . . 12
3.7 Ensemble . . . 14
3.8 Verdict . . . 15

4 TIC 120362137 (3+1) **15**
4.1 The system according to Borkovits et al. (2026) . . . 15
4.2 Stability . . . 16
4.3 A century forward and backward, and the apsidal motion . . . 16
4.4 TESS light curves . . . 18
4.5 Times of minima . . . 20
4.6 The fit . . . 21
4.7 Ensemble . . . 22
4.8 Verdict . . . 23

5 V994 Her ((2+2)+2) **23**
5.1 The system according to Zasche et al. (2023) . . . 23
5.2 Stability . . . 24
5.3 Thirty years forward and backward, and the apsidal motion . . . 24
5.4 TESS light curves . . . 26
5.5 Times of minima . . . 28
5.6 The fit . . . 28
5.7 Ensemble . . . 30
5.8 Verdict . . . 31

6 The three systems compared **32**

7 Discussion **33**
7.1 Published elements are not portable without anchoring 33
7.2 The dynamics is right; the classical apsidal motion tests stellar structure 33
7.3 What the fit can and cannot do 33
7.4 The role of the observers 34
7.5 Limitations of this test 34

8 Conclusions **34**

9 Data and reproducibility **35**

References **35**

Abstract. We test whether the N-body simulator SWINGERsim reproduces three well-studied compact multiple stars whose solutions were published by members and collaborators of the SQUADRA group: the 2+2 quadruple BU CMi (Pribulla et al. 2023), the 3+1 quadruple TIC 120362137 (Borkovits et al. 2026) and the triply eclipsing sextuple V994 Her (Zasche et al. 2023). For every system SWINGERsim integrated the published solution forward and backward in time, tested its stability, computed synthetic TESS light curves and the times of all eclipses, fitted its own N-body model to the measured times of minima, radial velocities and TESS light curves, and integrated an ensemble of 400–600 systems drawn within the published uncertainties.

All three systems are stable in every test. The published elements, entered literally, give eclipses that drift by hours to days within a few years, because osculating periods are not the periods with which eclipses repeat; after the period and phase of each eclipsing pair are adjusted to the minima (“anchored”), the synthetic TESS light curves agree with the observed ones to 2–4 parts per thousand (ppt) of the flux in the best sectors, about one per cent of the eclipse depth, and the eclipse times to the scatter of the data. A joint fit of all data, with a realistic error for every source of minima and the posterior sampled by MCMC, reproduces the published masses to 1–5 % (BU CMi, TIC 120362137) and the inclinations to 0.15°. It also shows three differences, each of which we traced as far as the published information allows. (1) In BU CMi the published apsidal periods agree to 1 % with SWINGERsim’s point-mass integration. When SWINGERsim adds the apsidal advance caused by the tidal and rotational bulges of the stars, the fit reproduces the observed apsidal motion by lowering the outer eccentricity by 2.6σ; with point masses the tension disappears. The published code can include tidal and rotational effects, and the paper’s own analytic estimate of the tidal advance is even somewhat larger than SWINGERsim’s; whether the published integration used a smaller tidal term or none is not stated, and remains open. (2) In TIC 120362137 the published apsidal period of the 51-d orbit is a theoretical value, not a measurement; SWINGERsim’s 32.4 yr is therefore not in conflict with the data. (3) In V994 Her the apsidal motion is mostly classical: SWINGERsim predicts it from stellar structure, while the paper fitted the rates freely. For pair B the paper itself finds an apsidal constant about 0.83 times the theoretical one, close to the factor 0.81 that SWINGERsim needs; pair C would need apsidal constants 1.8 times larger than stellar models give.

1 Aim of the test

SWINGERsim is a forward model: from masses, radii and orbits it computes how a hierarchical multiple star moves, what its light curve looks like and when its eclipses happen. SQUADRA works the other way round: it starts from observations, light curves and times of minima measured by satellites and by observers, and asks which system produced them. Before SWINGERsim can help with that, it has to pass a simple test: **given a system that has already been solved carefully by others, does SWINGERsim reproduce what was observed?**

We chose three systems with complete published solutions, long series of times of minima and TESS light curves, one of each hierarchy that matters for SQUADRA:

- **BU CMi**: two eclipsing binaries orbiting each other (2+2), the most compact known flat doubly eclipsing quadruple (outer period 121.8 d);

- **TIC 120362137**: an eclipsing binary with a third star on a 51-d orbit and a fourth on a 1046-d orbit (3+1), the most compact known quadruple of this type;
- **V994 Her**: three eclipsing binaries, A and B in a 2+2 and the visual companion C ((2+2)+2), the first known triply eclipsing sextuple.

For each system we ask five questions:

1. **Stability.** Is the published system dynamically stable, by analytic criteria and by a direct N-body test?
2. **Dynamics in time.** What happens to the orbits over a century forward and backward, and do the precession periods that SWINGERsim computes agree with the published ones?
3. **Light curves.** Do the synthetic light curves of SWINGERsim, of the whole system and of the individual pairs, match the TESS light curves?
4. **Times of minima.** Do the eclipses of SWINGERsim happen when the observers saw them, over the whole span of the published minima?
5. **Fit.** Can SWINGERsim's own fit to all these data find a configuration that reproduces them, and how does it compare with the published solution? An ensemble of systems within the published errors shows how robust the conclusions are.

Success means: the published system is stable in SWINGERsim; the synthetic light curves agree with TESS to the level of the residuals of the published models (a few ppt); the eclipse times agree with the measured minima to minutes once the periods are anchored; and the fit reproduces the data without leaving the range of physically sensible parameters. Where SWINGERsim disagrees with a paper, we look for the reason rather than for agreement.

2 Data and methods

2.1 The three systems

Table 1. The test systems and the data used. *Minima*: published times of minima (number, years, sources). *TESS*: sectors of the light-curve comparison (SPOC: pipeline of the TESS Science Processing Operations Center; QLP: Quick-Look Pipeline).

	BU CMi	TIC 120362137	V994 Her
hierarchy	2+2: (Aa+Ab)+(Ba+Bb)	3+1: ((Aa+Ab)+B)+C	(2+2)+2: ((A)+(B))+(C)
eclipsing pairs	A 2.94 d, B 3.26 d	Aa–Ab 3.28 d	A 2.08 d, B 1.42 d, C 1.96 d
outer orbits	A–B 121.8 d	A–B 51.3 d, AB–C 1046 d	A–B 1062 d; C around AB (not measured)
source of the solution	Pribulla et al. (2023), Table 7	Borkovits et al. (2026), Tables 1–2	Zasche et al. (2023), Table 3
epoch of the elements (BJD)	2458491.5	2458683.3	2459010.0
minima	183, 2005–2023; HAT, KELT, MASCARA, TESS, ground follow-up, Ogloza et al. (2008), Volkov et al. (2021)	147, 2019–2024; Supplementary Table S3 / Supplementary Data 6 of the paper	343, 2002–2022; TESS, observers RU, SD & UM, FRAM, MM, GP
radial velocities	305 of all four stars (Pribulla et al. 2023)	380 of all four stars (Borkovits et al. 2026)	128 of the stars of A and B (Lee et al. 2008)
TESS sectors	7 (SPOC 30 min), 34, 61 (SPOC 2 min)	14, 26, 40–41, 53–54, 74 (SPOC)	26 (SPOC 2 min), 40, 53 (QLP 10 min)

The published solutions of all three systems were obtained with the code Lightcurvefactory (Borkovits et al. 2019, 2020, 2021), which fits the light curves, the radial velocities, the eclipse timing variations and the spectral energy distribution together, with the radii and temperatures of the stars taken from stellar isochrones (PARSEC). The orbits are computed in two different ways. For BU CMi and TIC 120362137 the code integrates the motion of the stars numerically (a *photodynamical* model); its integrator can include tidal and rotational effects (Borkovits et al. 2019, Appendix A), but neither paper states whether these terms were used. For V994 Her the perturbations between the pairs are small, and Zasche et al. (2023, Sect. 3) used analytic Keplerian orbits with a constant, freely fitted apsidal advance rate for each pair instead of an N-body integration. The orbital elements in the papers are *instantaneous osculating elements* at the epoch in Table 1, i.e. the Keplerian orbit that the stars would follow from that moment if the other stars disappeared. SWINGERsim uses the same kind of elements as initial conditions, with the argument of periastron ω of the secondary (relative orbit), the convention of all three papers. For V994 Her the papers give no nodes; we assumed Ω = 0 for all orbits (the authors consider the system close to coplanar) and an orbit of C around A+B with P = 3000 yr and e = 0.3 (the projected separation is ∼300 au); the ensemble varies these assumptions.

2.2 Terms used in this report

- **Time of minimum (eclipse time).** The moment of the deepest light loss in an eclipse, measured by an observer from a light curve: these are **data**. The papers list them with errors, the *primary* minimum being the eclipse of the first star of a pair.
- **Model eclipse time.** For every measured minimum SWINGERsim computes the moment of minimum projected separation of the two stars of the pair in the N-body integration, plus the time the light needs to cross the system to the observer (Irwin 1952). It is the model's version of the same event, not a prediction.
- **O − C** ("observed minus calculated"). The measured time minus the time of a strictly periodic (linear) ephemeris T = T0 + P E, where E is the cycle number. A strictly periodic binary gives a horizontal line. In a multiple star the line bends: the *apsidal motion* moves the primary and secondary minima in opposite directions (two mirror-image waves); the *light-travel-time effect* (LTTE) of an outer orbit shifts all minima of a pair together; and in compact systems the third star changes the period directly (*dynamical* eclipse timing variations, ETV). In this report the linear ephemeris of each pair is the mean ephemeris of the model primaries, so that measured points and model curves share one reference.
- **Prediction.** A model eclipse time or light curve at a date without data.
- **Osculating and mean period.** The osculating period is that of the Keplerian orbit at one instant; the mean period is the average interval between eclipses. In a compact multiple star they differ by 10^{-4}–10^{-3} of the period, because the orbit of a pair is perturbed continuously.
- **Anchoring.** Adjusting only the period and the phase (mean anomaly at the epoch) of each eclipsing pair so that its mean eclipse period and phase match the minima.
- **Apsidal motion.** Rotation of the line of apsides (the long axis) of an eccentric orbit; its period U is the time for one full turn. It is driven by the other stars (*dynamical*), by the tidal and rotational bulges of the stars themselves (*classical*), and by general relativity.
- **Stability ratio.** For a pair inside an outer orbit, the ratio of the outer periastron distance to the critical distance of Mardling & Aarseth (2001, "MA01") or Vynatheya et al. (2022, "V22"); above 1 the triple is stable. For quadruples SWINGERsim also uses the machine-learning classifier of Vynatheya et al. (2023) and a *ghost test*: two N-body runs differing by 10^{-6} au in one semi-major axis must stay close.
- **Extra light.** Light in the photometric aperture that does not take part in the eclipses: other stars of the system, neighbouring stars, or light removed or added by the pipeline's correction for crowding. It changes all eclipse depths by the same factor.
- **ppt**: parts per thousand of the total flux. **rms**: root mean square of the residuals. **Median |O − C|**: the typical residual, less sensitive to a few bad points.

2.3 Three model configurations

Every system was computed with three configurations:

- **published**: exactly the elements, masses, radii and temperatures of the paper;
- **anchored**: the published configuration with the period and phase of each eclipsing pair adjusted to the minima; all other numbers as published;
- **fitted**: the result of SWINGERsim's fit (below).

Why anchoring is needed is one of the results of the test: osculating periods are not portable. Two codes that describe the same motion with slightly different coordinates or forces map the same osculating numbers to slightly different mean periods, and over a few thousand cycles a difference of 10^{-4} in the period becomes hours. Anchoring removes this difference with two numbers per pair and leaves the dynamics untouched.

2.4 The model

SWINGERsim integrates the motion of all stars with full gravity (REBOUND with the IAS15 integrator, Rein & Liu 2012, Rein & Spiegel 2015), including general relativity and the tidal and rotational deformation of the stars (REBOUNDx, Tamayo et al. 2020), with the apsidal constants from stellar models (Claret 2023) at the age given by MIST tracks (Choi et al. 2016; Dotter 2016). The light curve sums the light of all stars: limb-darkened spheres (or Roche surfaces) with the four-coefficient law of Claret (2000) and coefficients of Claret & Bloemen (2011), with the eclipses of every star by every other star in front of it, the ellipsoidal variation, reflection and Doppler beaming of the close pairs, and the light-travel time across the system.

2.5 Comparing light curves

The light curves of the TESS satellite (Ricker et al. 2015) were downloaded with lightkurve (Lightkurve Collaboration 2018) from the SPOC (Jenkins et al. 2016) and QLP (Huang et al. 2020) pipelines. The model was computed at the observed times and averaged over the exposure (2, 10 or 30 min). Two things in the TESS photometry are not properties of the stars and have to be allowed for: extra light in the aperture changes all depths by the same factor, and slow instrumental trends bend the baseline. We therefore compared the observations with `a × model + baseline`, where `a` is one scale factor per sector and the baseline is a quadratic polynomial in time for each continuous segment. These are the only adjusted numbers; the eclipses come from the model unchanged. The figure of merit is the rms of observed minus model in ppt, after removing points more than five robust standard deviations away.

2.6 The fit

The fit varies the elements of the N-body model until it matches the data. For every measured minimum the system is integrated to just before the expected eclipse, and the model eclipse is found from the motion of the pair. The fit minimises $\chi^2 = \Sigma\,((\text{data} - \text{model})/\sigma)^2$ over all data at once:

- **times of minima**, every source (each survey, each observer, each TESS sector) with its own extra error ("jitter"), added in quadrature to the quoted errors and determined by the fit so that every source has a reduced χ^2 of 1; minima more than 5 robust standard deviations from the model are left out (and taken back if a later iteration brings them closer);
- **radial velocities**, with the masses of the stars, the systemic velocity and a zero point per instrument free;
- **light curves**: windows of ±0.3 d around the eclipses of one or two TESS sectors, with the inclinations and radii free and a linear calibration per window (scale for extra light, zero point, trend). Neighbouring points of a light curve are not independent (red noise); their errors are multiplied by the factor β of Pont et al. (2006), measured from the residuals;
- **priors**: Gaussian priors from the papers on the elements that the data constrain only weakly (eccentricities and ω of the pairs, the outer orbits), never on the periods of the pairs (osculating periods are not portable).

The minimisation uses a trust-region least-squares method (SciPy, Virtanen et al. 2020). The posterior was then sampled with emcee (Foreman-Mackey et al. 2013; 72 walkers × 1000

steps, the first half discarded as burn-in). The autocorrelation time of the chains is $\tau \approx 52$–53 steps, so the whole chain covers about 19 τ and the kept second half about ten τ per walker. That meets the rule of thumb of Foreman-Mackey et al. (2013), who suggest running for about ten τ, and is enough for the 16–84 % ranges quoted here (about 700 independent samples), but it is short of the 50 τ that the emcee online documentation (tutorial "Autocorrelation analysis & convergence") recommends for a reliable estimate of τ itself. A run of 400 steps gave errors that differ from those of 1000 steps by a median of 3–5 % (at most a factor of 2, for the eccentricity vector of the almost circular pair of TIC 120362137) and medians that differ by less than 0.4σ. The quoted errors are therefore reliable to a few tens of percent.

To see what each kind of data contributes, the fit was also run step by step: times of minima alone, then with the extra error per source, with priors, starting automatically from the published elements, with radial velocities, and with light curves. The steps are summarised for each system.

2.7 Ensembles

An ensemble is a set of systems drawn at random within the published uncertainties (masses, eccentricities, inclinations, nodes, the outer orbits; for V994 Her also the unknown orbit of C). Each system is integrated forward and backward, and SWINGERsim records its fate (intact, unstable, merged), its precession periods and whether its pairs eclipse. Systems are then weighted by how well they reproduce the published apsidal periods and the condition that the pairs eclipse now. The effective number of members n_eff measures how many systems carry the weight.

3 BU CMi (2+2)

3.1 The system according to Pribulla et al. (2023)

BU CMi consists of two eclipsing binaries of A-type stars, A (2.94 d) and B (3.26 d), each with an eccentric orbit ($e \approx 0.22$), which revolve around each other in only 121.8 days. All three orbits lie within one degree of each other; the four stars have 2.3–2.5 solar masses and an age of about 260 Myr. The short outer period makes BU CMi a showcase of fast dynamical apsidal motion: both pairs turn their orbits once in 25–29 years, and the eclipse times swing by hours. The authors determined everything from TESS light curves, radial velocities of all four stars, 183 times of minima since 2005 and the spectral energy distribution (Pribulla et al. 2023, Table 7).

3.2 Stability

Table 2. Stability of the published BU CMi. A ratio above 1 means stable.

sub-triple	a_in / a_out	e_in	e_out	i_mut	MA01	V22
A inside the A–B orbit (+B)	0.066	0.219	0.273	0.4°	2.53	2.89
B inside the A–B orbit (+A)	0.071	0.226	0.273	0.2°	2.36	2.69

The machine-learning classifier of Vynatheya et al. (2023) for 2+2 quadruples gives *stable*, and the N-body ghost test over 100 outer orbits (33 years) ends with a relative divergence of 1.2×10^{-3}, far below the threshold of instability. BU CMi is compact (the outer periastron is only about 2.4 times the critical distance) but clearly stable.

3.3 A century forward and backward, and the apsidal motion

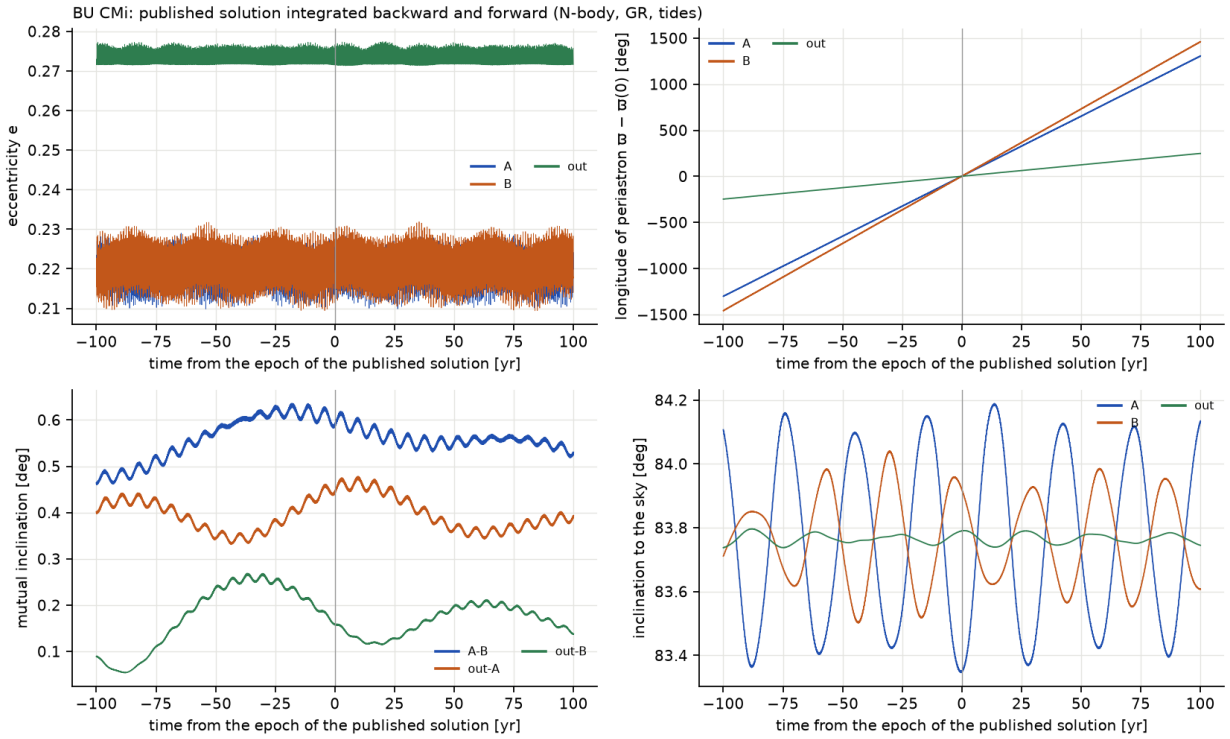


Figure 1. BU CMi: the published solution integrated 100 years backward and forward. Top left: eccentricities of the three orbits; top right: longitude of periastron relative to its value at the epoch (the apsidal motion); bottom left: mutual inclinations; bottom right: inclinations of the three orbits to the sky.

How to read the figure. The horizontal axis is time in years from the epoch of the published elements (BJD 2458491.5, January 2019); left of zero is the past, computed by integrating backward. Each panel shows one property of the three orbits (pair A, pair B and the outer orbit "out"). Top right: a straight rising line is an orbit whose periastron turns at a constant rate; its slope is 360° divided by the apsidal period. Bottom right: a pair eclipses only while its inclination stays close to 90°.

The system stays intact in both directions, the relative energy error stays below 2.3×10^{-11}, the eccentricities oscillate within 0.210–0.229 (A), 0.209–0.232 (B) and 0.271–0.277 (outer), and both pairs eclipse during the whole two centuries. The mutual inclinations stay small: A–B 0.52–0.62° (published 0.61°), outer–A 0.34–0.48° (0.45°), outer–B 0.11–0.21° (0.26°).

Table 3. Apsidal periods of BU CMi. The published values were measured from the authors' numerical integration; SWINGERsim measures them from its own, with and without the tidal and rotational bulges of the stars.

	published (from the authors' integration)	SWINGERsim: point masses + relativity	SWINGERsim: + tides and rotation	ensemble with tides, weighted (16–84 %)
pair A	28.7 ± 0.1 yr	28.98 yr (+1.0 %)	27.61 yr (−3.8 %)	27.7 (26.8–28.5) yr
pair B	25.1 ± 0.1 yr	25.30 yr (+0.8 %)	24.65 yr (−1.8 %)	24.6 (23.9–25.4) yr
outer orbit	145.3 yr	145.08 yr (−0.15 %)	145.07 yr	–

The fast precession is caused by the other pair: with point masses SWINGERsim reproduces

the published apsidal periods to 1 %, and the outer orbit to 0.15 %. The bulges of the stars add an advance of 0.56°/yr for A (about 16″ per orbit) and 0.33°/yr for B, which shortens the periods by 4 % and 2 %. This is the classical apsidal motion that the stellar structure predicts.

Pribulla et al. (2023, Table 7) estimate the same classical term analytically: Δω_tide = 18.73″ and 12.67″ per orbit. With the mean eclipse periods of the pairs (2.93959 and 3.26222 d, i.e. 124.3 and 112.0 orbits per year) this is **0.65°/yr for A and 0.39°/yr for B**, 15–20 % more than SWINGERsim's 0.56 and 0.33°/yr. The paper also lists apsidal periods from the analytic formula (31.75 and 28.96 yr) and, from a separate fit of the spectra and the TESS light curves of sectors 7 and 34, fitted apsidal periods of 24.91 ± 0.04 and 25.79 ± 0.08 yr (its Table 5). The values in the first column of Table 3 are different: they were measured from a numerical integration of the final photodynamical solution.

What does the comparison tell us about that integration? Lightcurvefactory can integrate the orbits with tidal and rotational effects (Borkovits et al. 2019, Appendix A), and the paper's analytic tidal advance is even larger than SWINGERsim's. If the published integration had contained a classical term of that size, SWINGERsim should match the published apsidal periods only when it includes its own tidal and rotational term. It does the opposite: the published periods agree with SWINGERsim's point-mass run to 1 % and differ from the run with bulges by 2–4 %. Either the tidal term in the published integration is smaller than both estimates (for example because of different apsidal constants k_2 or radii), or it was switched off. The paper does not say which, and we cannot decide it from the published numbers. This matters for the fit below.

3.4 TESS light curves

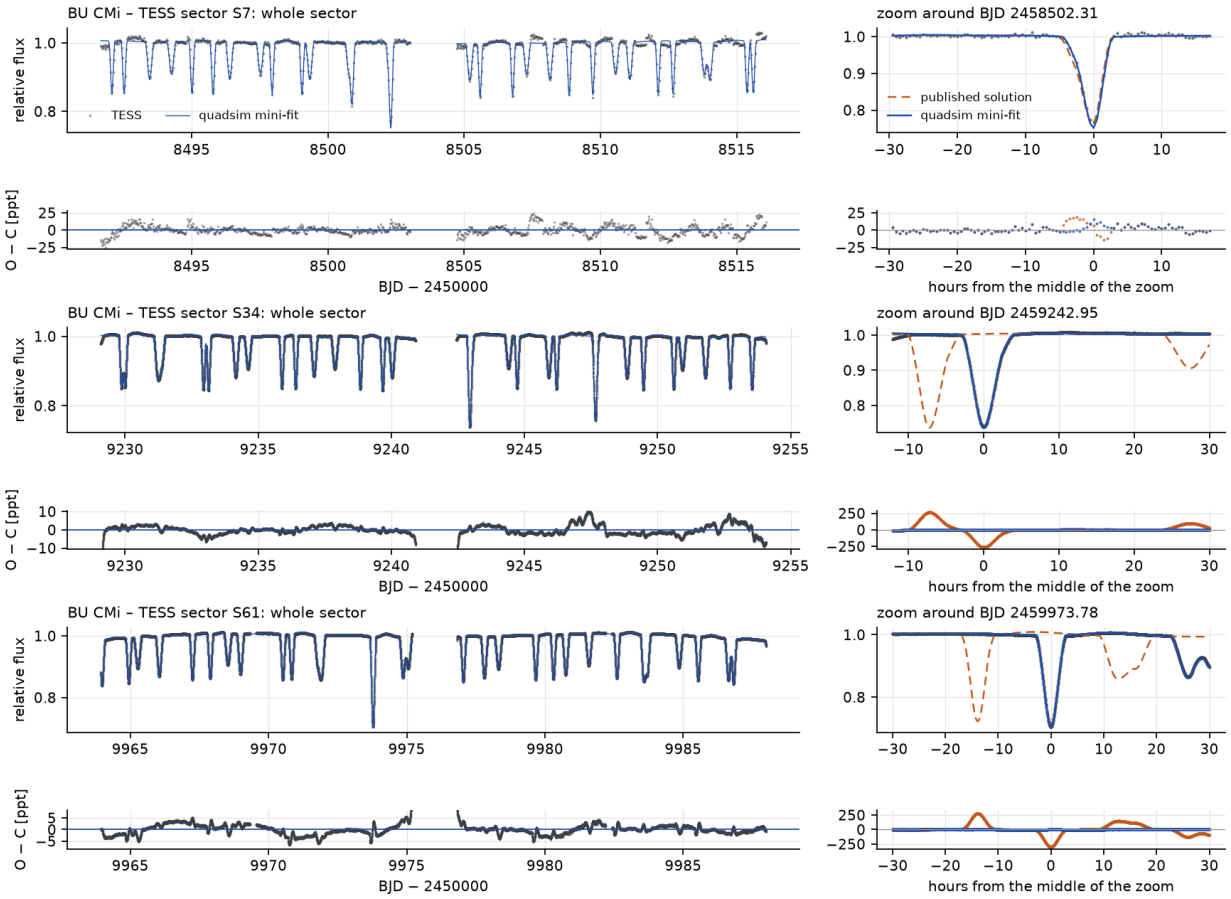


Figure 2. BU CMi: TESS light curves (grey points) and the SWINGERsim model (blue: fitted to the times of minima) for sectors 7, 34 and 61. Left: the whole sector with the residuals (observed − model, in ppt) below. Right: a 2.5-day zoom around the deepest eclipse in the middle of the sector, with the model of the published elements (orange, dashed) and of the fit (blue).

How to read the figure. Each pair of rows is one TESS sector. In the large panel every dip is an eclipse of pair A or B (the deeper narrow dips around 0.75–0.8 of the flux are those where an eclipse of A and of B overlap). If the model were perfect, the residual panel below would be a flat band of noise. In the zoom the time axis is in hours, so that the shape of one eclipse can be compared: the orange dashed curve shows where the eclipses of the *published* elements fall: in sector 7, the epoch of the elements, almost right; by sector 61 (four years later) many hours off. The blue curve (fit) lies on the data.

Table 4. Residuals of the TESS light curves of BU CMi (rms of observed − model, ppt). Eclipse depths are 150–250 ppt.

sector (cadence)	points	published	anchored	fitted to the minima	scale a
7 (30 min)	1 086	9.8	7.0	7.1	0.98
34 (2 min)	16 823	25.5	3.1	2.7	0.98
61 (2 min)	15 774	27.9	2.8	2.0	0.98

The published elements reproduce sector 7 (the epoch of the elements) but miss sectors 34 and 61 by 25–28 ppt, because their eclipses fall at the wrong times. Once the periods and phases are anchored, the light curves of all three sectors agree to 2–3 ppt in the two-minute

data, about one per cent of the eclipse depth. The scale factor a is fitted freely for each sector; its value of 0.98 means that the depths that follow from the published radii and temperatures are right to 2 %. The remaining residuals are slow waves of a few ppt, partly instrumental. Sector 7 has 30-min sampling, which blurs the eclipses; its larger rms reflects the coarse data, not the model.

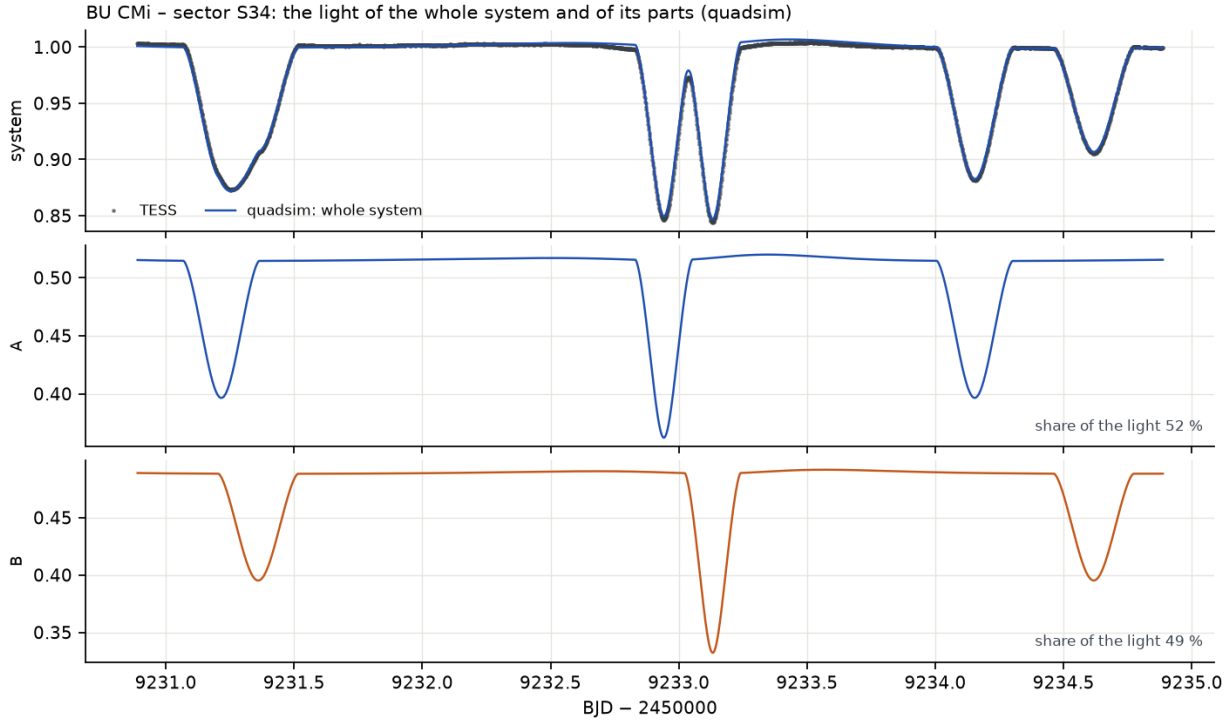


Figure 3. BU CMi, sector 34: the light of the whole system (top, with the TESS points) and, separately, of pair A and pair B as computed by SWINGERsim. Each pair contributes about half of the light.

How to read the figure. The top panel repeats four days of the TESS light curve with the model. The two panels below show what SWINGERsim knows and the telescope cannot see: the light of each pair alone. Every dip in the top panel can be traced to one of the pairs below, and where both pairs dip at the same time the system shows a double eclipse.

3.5 Times of minima

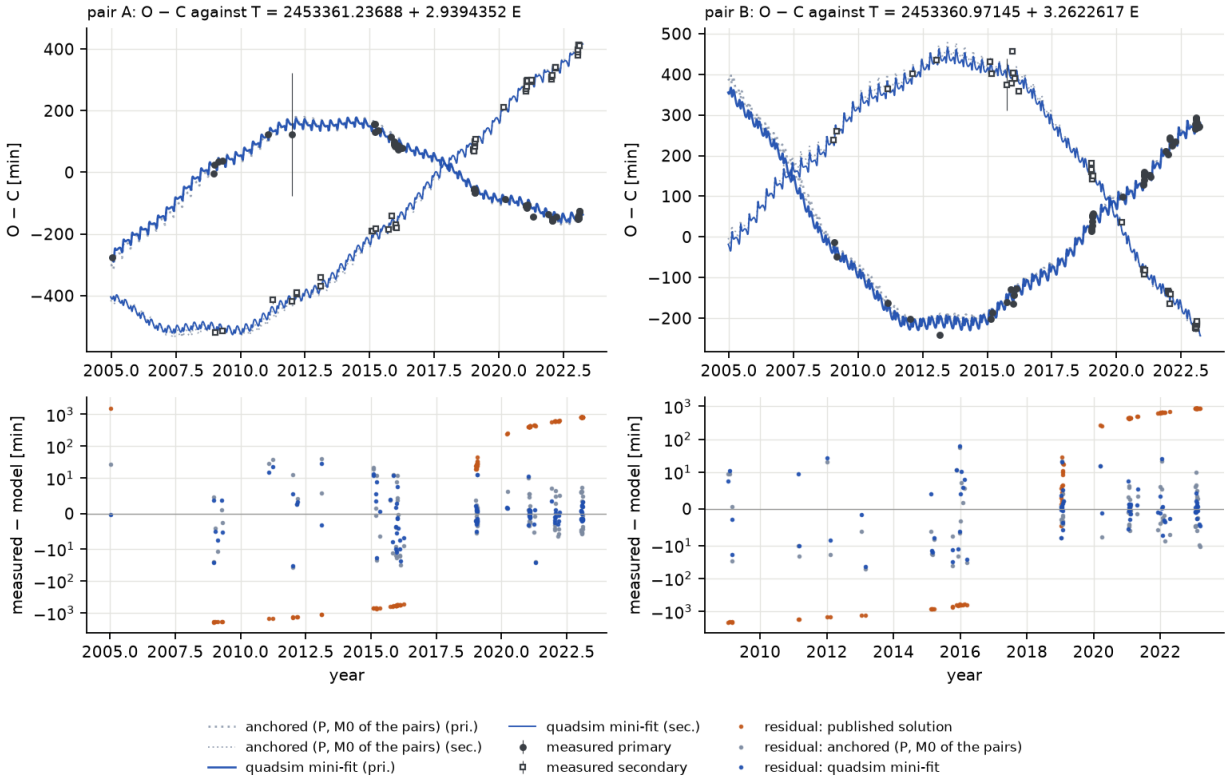


Figure 4. BU CMi: O − C diagrams of pair A (left) and pair B (right). Top: measured primary (filled) and secondary (open) minima and the models; bottom: measured − model residuals (symmetric logarithmic axis) for the published (orange), anchored (grey) and fitted (blue) configurations.

How to read the figure. Each point is one measured eclipse; its height is how many minutes earlier or later it happened than a strictly periodic ephemeris predicts. The two mirror-image waves of each pair, primaries and secondaries moving in opposite directions over about 28 years, are the apsidal motion; the small fast ripples on the curves are the 121.8-day outer orbit (light time and perturbations by the other pair). The lower panels have a logarithmic scale: the orange points of the published elements lie at hundreds to thousands of minutes, the anchored and fitted ones within ±10 minutes for most minima.

With the published elements the eclipses drift by up to 32 hours (A) and 35 hours (B) over 18 years: the osculating periods of the paper correspond in SWINGERsim to mean eclipse periods shorter by 3.3×10^{-4} (A) and 4.0×10^{-4} (B). Anchoring two numbers per pair reduces the rms to 11–13 minutes, and the shape of the O − C (amplitude and phase of the apsidal motion and the 121.8-d ripples) is reproduced without any further adjustment.

Table 5. The minima of BU CMi by source: quoted errors and the extra error that the fit assigns to each source (Pribulla et al. 2023, Tables 3–4). The six sources below give 182 of the 183 minima; the remaining one is a single minimum of January 2005 from Ogloza et al. (2008), the oldest point of the data set; one point cannot measure a scatter, so the fit adds no extra error to it. Volkov et al. (2021) give no errors for their minima; the 9 s is the error of 0.0001 d that Pribulla et al. (2023) assigned to them.

source	years	minima	median quoted σ	extra error found by the fit
HAT	2008–2009	10	55 s	11.5 min
KELT	2011–2013	13	118 s	18.2 min
Volkov et al. (2021)	2012–2021	8	9 s	11.1 min

source	years	minima	median quoted σ	extra error found by the fit
MASCARA	2015–2016	39	41 s	10.3 min
TESS	2019–2023	87	6 s	3.4 min
ground follow-up	2021–2023	25	11 s	2.8 min

The survey minima (HAT, KELT, MASCARA) scatter by 10–18 min around any model, ten to twenty times their quoted errors; TESS and the recent ground-based follow-up need only about 3 min. With these extra errors the old minima still define the long-term apsidal motion but no longer pull the solution.

3.6 The fit

Table 6. BU CMi: what each kind of data adds. *rms*: of the minima used, pairs A / B; e_out and ω_out: the outer orbit (published 0.2728 ± 0.0037 and 141.0 ± 3.6°).

fit	data	χ^2_red	rms A / B [min]	e_out	ω_out [°]
minima alone, quoted errors	179 minima	1432	6.6 / 13.0	0.261 ± 0.003	131 ± 2
+ extra error per source	176	0.98	6.9 / 8.8	0.261 ± 0.003	130 ± 2
+ priors	177	0.99	8.3 / 9.4	0.262 ± 0.002	135 ± 2
automatically from the published elements	176	0.99	8.4 / 8.9	0.261 ± 0.002	136 ± 2
+ radial velocities	176 + 305 RV	1.00	7.8 / 9.3	0.263 ± 0.002	136 ± 2
+ light curves, red noise, MCMC (final)	177 + 305 RV + 4147 LC	–	7.6 / 9.1	0.263 ± 0.002	132 ± 1
point masses + radial velocities	176 + 305 RV	1.00	8.7 / 10.3	**0.271 ± 0.002**	**136 ± 2**

Table 7. BU CMi: published elements (Pribulla et al. 2023, Table 7) and the final SWINGERsim fit (minima, radial velocities, TESS sectors 34 and 61; posterior median and half the 16–84 % range).

quantity	published	SWINGERsim	difference
e_A	0.2191 ± 0.0008	0.2152 ± 0.0005	−4.9σ (prior)
ω_A [°]	101.41 ± 0.13	101.75 ± 0.13	+1.9σ
e_B	0.2257 ± 0.0009	0.2234 ± 0.0006	−2.6σ (prior)
ω_B [°]	257.52 ± 0.15	257.21 ± 0.12	−1.6σ
P_out [d]	121.793 ± 0.031	121.717 ± 0.015	−2.5σ
e_out	0.2728 ± 0.0037	0.2631 ± 0.0017	−2.6σ
ω_out [°]	141.0 ± 3.6	132.2 ± 1.4	−2.4σ
masses [$M_\odot$]	2.536, 2.384, 2.521, 2.320	2.51 ± 0.09, 2.33 ± 0.11, 2.55 ± 0.12, 2.20 ± 0.09	−1 %, −2 %, +1 %, −5 %
inclinations A, B [°]	83.35, 83.92	83.34 ± 0.10, 83.82 ± 0.09	−0.01°, −0.10°
radii [$R_\odot$]	2.249, 2.072, 2.235, 2.003	2.32 ± 0.06, 1.98 ± 0.07, 2.30 ± 0.07, 1.91 ± 0.08	sums within 0.5 %

quantity	published	SWINGERsim	difference
TESS light curve rms	–	3.1 ppt (S34), 5.8 ppt (S61)	scale a = 0.985, 0.995

The masses and inclinations agree with the paper within their errors. The radii show the familiar degeneracy of partial eclipses: the sum of the radii of each pair is fixed by the duration of the eclipses and agrees with the paper to 0.5 %, but the ratio is weakly determined by the light curves. In the paper the individual radii follow from the joint model of the light curves, the spectral energy distribution and the PARSEC isochrones (Pribulla et al. 2023, Table 7, where the temperatures, luminosities and gravities are marked as interpolated from PARSEC); our fit determines the radii from the light curves alone, without the spectral energy distribution or isochrones.

The outer orbit and the classical apsidal motion. In every fit that includes the tidal and rotational bulges of the stars, the outer eccentricity is about 0.263, 2.6σ below the published value, and e_A is pulled below its prior. When the same fit is done with point masses, e_out = 0.271 ± 0.002 and all priors are satisfied within 1.4σ. The explanation follows from Table 3: the measured apsidal waves in the O − C diagram fix the total apsidal rate. With the classical term included, the dynamical part must be smaller, and the fit makes it smaller by lowering the outer eccentricity. Both descriptions fit the minima equally well. What is established is this: SWINGERsim's point-mass integration reproduces the published apsidal periods to 1 %, and adding SWINGERsim's tidal and rotational term moves the fitted e_out by 2.6σ. What is open is why the published solution does not need this shift. Both SWINGERsim (0.56 and 0.33°/yr) and the paper's own analytic estimate (0.65 and 0.39°/yr) predict a classical advance of this size; if the published integration contained a smaller tidal term or none, its outer eccentricity absorbed the difference, and the true value would be closer to 0.263 than to 0.273. The paper does not state which tidal model its integration used. The distinction would show in the next decade of minima, because the two models predict slightly different apsidal waves.

The apsidal constants set free. The same fit with the apsidal constant of one star of each pair free (stars m1 and m3; their partners keep the model values; models: 0.0042 for both) settles at k_2 = 0.0003 for pair A (the lower bound) and 0.0045 ± 0.0021 for pair B. The outer eccentricity moves only from 0.2631 to 0.2663 ± 0.0021, a third of the way to the published value. Because a value at a bound has no meaningful linear error, the constraint comes from a χ^2 profile: seven fits with k_2 of m1 fixed at 0.0003–0.009 and everything else free, all with the errors of the free fit held fixed so that their χ^2 can be compared (**Table 3a**).

Table 3a. BU CMi: χ^2 profile in the apsidal constant of star m1 (pair A).

k_2 (m1)	0.0003	0.0010	0.0020	0.0030	0.0042	0.0060	0.0090
$\Delta\chi^2$	0	0.70	2.98	6.73	13.54	27.95	53.00
k_2 (m3), fitted	0.0044	0.0034	0.0020	0.0006	0.0003*	0.0003*	0.0003*
$e_{\rm out}$	0.2659	0.2658	0.2656	0.2655	0.2647	0.2637	0.2624

* at the lower bound.

The data accept $k_2 \leq 0.0012$ for pair A at $\Delta\chi^2$ = 1 and ≤ 0.0023 at $\Delta\chi^2$ = 4. The model value 0.0042 costs $\Delta\chi^2$ = +13.5 even with the constant of pair B pushed to its lower bound. Up to $k_{2,A}$ = 0.003 the two pairs trade off ($k_{2,A} + k_{2,B} \approx 0.0045$), so the minima measure mainly their sum, about half of what the models give. The $\Delta\chi^2$ values are conservative, because the frozen errors include the red-noise factor of the light curves (χ^2_red = 0.29). The tidal and rotational advance the data want is therefore weaker than that of synchronously rotating stars with the model constants. Lower k_2, sub-synchronous rotation (which lowers the rotational part) or physics missing from the model would each explain it. The fit cannot tell them apart, and freeing the constants does not explain the outer eccentricity of the paper.

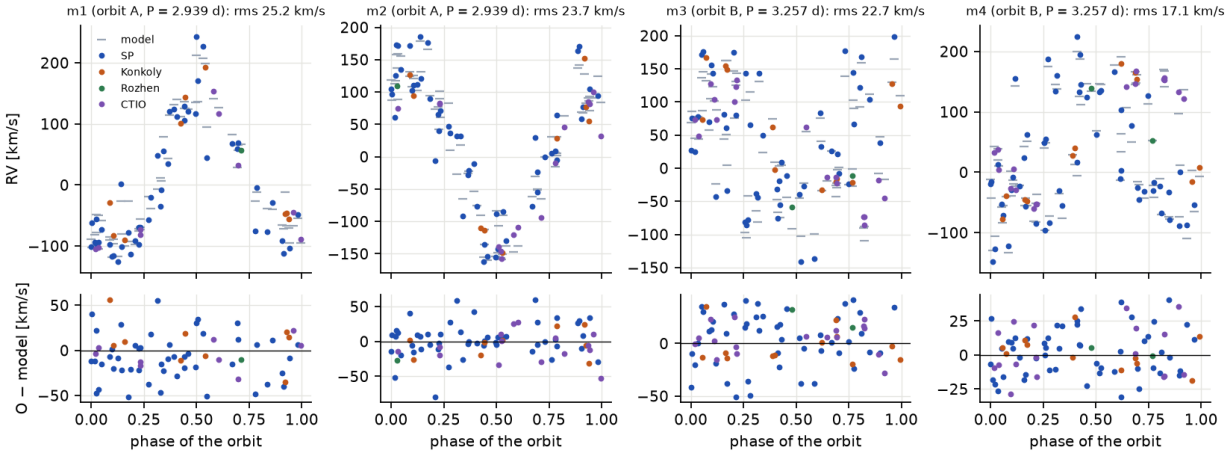


Figure 5. BU CMi: radial velocities of the four stars against the phase of their pair, from a fit of minima, velocities and light curves, with the model at the times of the observations (grey ticks) and the residuals.

How to read the figure. Each column is one star; the model ticks scatter around a sine curve because the pairs move on the 121.8-d outer orbit. The residuals of 17–25 km/s (rms) are seven to ten times the quoted errors of 2–3 km/s and show no clear pattern with phase or instrument.

The radial velocities scatter by 17–25 km/s around any model, also around a free Keplerian fit of each pair with a linear drift of ω (24 and 20 km/s rms), and swapping the components does not reduce it. The velocities were measured from broadening functions of four similar A stars whose lines blend at most phases; Pribulla et al. (2023) note the blending and the difficulty of identifying the components and quote no rms. The scatter is therefore a property of the velocities, not of the orbits; the fit gives each instrument an extra error of 16–19 km/s, and the masses come out within 1–5 % of the published ones.

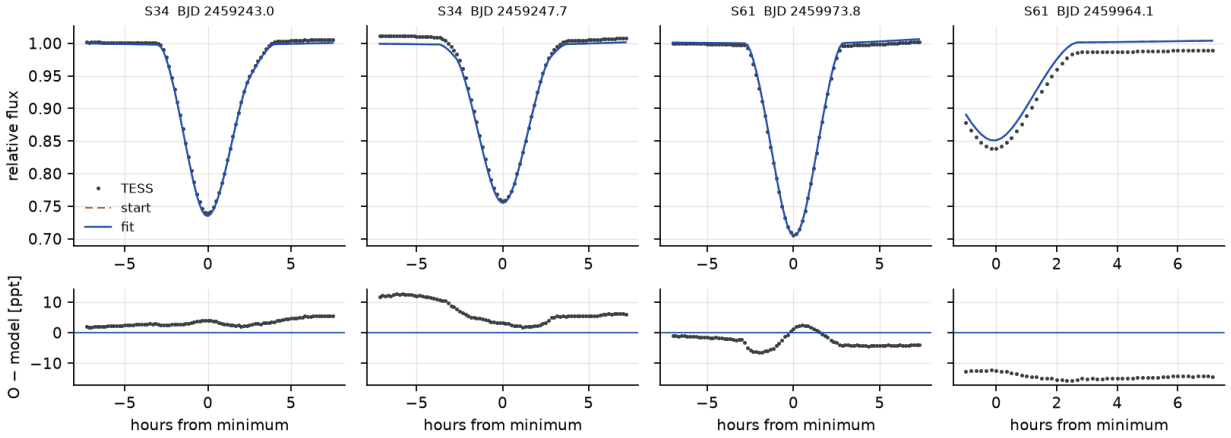


Figure 6. BU CMi: four eclipses of the light-curve fit – TESS (points), start model (orange, dashed), fit (blue) – and the residuals.

How to read the figure. Each column is one window of ±0.3 d around an eclipse, the time axis in hours from the model minimum. The fit and the start model are almost identical: the anchored configuration already matched the light curve (Table 4).

3.7 Ensemble

600 systems were drawn within the published errors of the masses, eccentricities, inclinations, nodes and the outer orbit, and integrated 40 years forward and backward. **All**

600 stay intact in both directions and are stable by MA01 and V22; both pairs eclipse in all of them. Weighted by the published apsidal periods (Gaussian, σ = 0.5 yr) and by eclipses of both pairs, n_eff = 238.

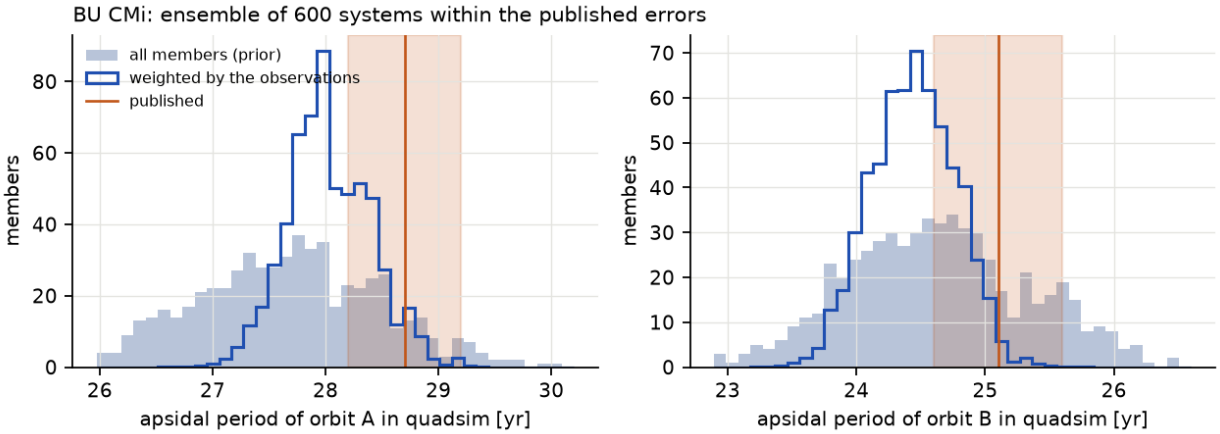


Figure 7. BU CMi: distribution of the apsidal periods of pair A (left) and B (right) in the ensemble – all members (filled) and weighted by the observations (outline) – with the published values and their range (orange).

How to read the figure. Each histogram counts how many systems of the ensemble have a given apsidal period. The filled histogram is what the published uncertainties allow; the outlined one gives more weight to systems close to the published apsidal periods. The weighted distributions peak 1 year below the published values: within the published uncertainties of masses and orbits, SWINGERsim's apsidal motion with tides is systematically 2–4 % faster, the classical term of Table 3.

3.8 Verdict

- **Confirms** the published solution: stable, the dynamical apsidal motion reproduced to 1 % with point masses, the light curves to 2–3 ppt, the 18-year O − C of both pairs with their apsidal waves and the 121.8-d ripples, the masses to 1–5 % and the inclinations to 0.1°.
- **Adds:** realistic errors for every source of minima (surveys 10–18 min, TESS and follow-up 3 min).
- **Differs, cause partly open:** with the classical apsidal advance that stellar models predict, the timings require an outer eccentricity of 0.263 instead of 0.273 (2.6σ); with point masses they agree with the paper. The paper's own analytic tidal advance is of the same size, so the difference points to the tidal term in the published integration, which the paper does not describe. With the apsidal constants free the minima want a weaker tidal and rotational advance than the models give ($k_2 \leq 0.0023$ for pair A at $\Delta\chi^2 = 4$, against 0.0042), and e_out still stays at 0.266. Projected rotational velocities would separate the rotational part.
- **Data to check:** the radial velocities scatter by 17–25 km/s around any model, far beyond their quoted errors.

4 TIC 120362137 (3+1)

4.1 The system according to Borkovits et al. (2026)

TIC 120362137 is the most compact known quadruple of the 3+1 type: an eclipsing pair of F stars (Aa–Ab, 3.28 d, nearly circular), a third F star (B) orbiting the pair in 51.3 days, and a G star (C) orbiting the triple in 1046 days. The 51-d orbit is so tight that B causes strong dynamical eclipse timing variations and occasional *third-body eclipses*, when B passes in front of or behind the inner pair. Only Aa–Ab eclipses regularly; everything about B and C

is learnt from the timing of these eclipses, the third-body events, radial velocities and the spectral energy distribution. A bright neighbour 27″ away contributes about half of the light in the TESS aperture: the paper's Table 2 gives an extra light of ℓx(TESS) = 0.510 ± 0.007.

4.2 Stability

Table 8. Stability of the published TIC 120362137.

sub-triple	a_in / a_out	e_in	e_out	i_mut	MA01	V22
A–B orbit inside AB–C (+C)	0.126	0.224	0.274	1.3°	1.63	1.85
Aa–Ab inside A–B (+B)	0.141	0.007	0.224	0.4°	1.48	1.86

Both ratios are well above 1 but lower than in BU CMi: TIC 120362137 is more compact relative to its stability limit, although Borkovits et al. (2026) find the stability conditions "fulfilled very robustly", and our ratios agree with that. The classifier of Vynatheya et al. (2023) gives *stable*, and the ghost test over 100 outer orbits (286 years) ends with a divergence of 9.9×10^{-3}, larger than in BU CMi, as expected for such a compact system, but still stable.

4.3 A century forward and backward, and the apsidal motion

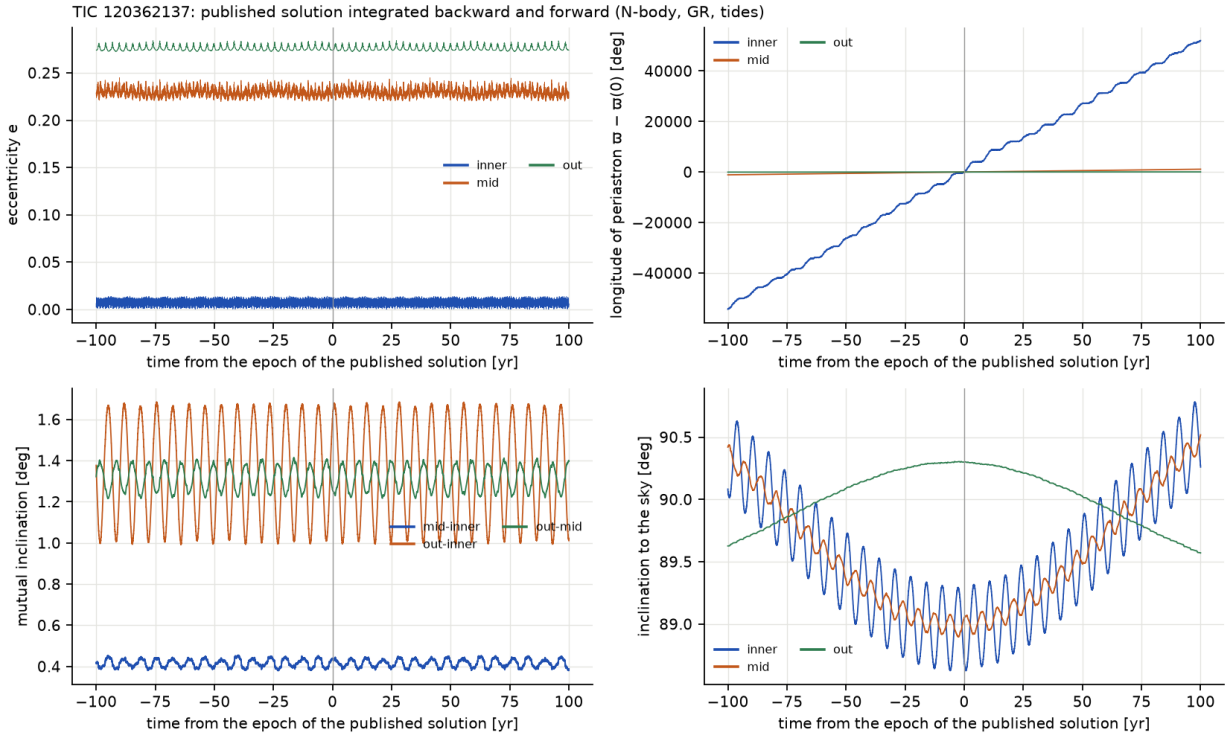


Figure 8. TIC 120362137: the published solution integrated 100 years backward and forward; panels as for BU CMi.

How to read the figure. As for BU CMi. The eccentricity of the inner pair (top left, blue) oscillates between almost zero and 0.014 within weeks: it is *forced* by the third star at every passage. Bottom right: the inclinations of the inner and middle orbits to the sky oscillate by up to ±0.8° on time scales of years to a century (the nodal precession of the orbits around the plane of the whole system); this is why the depths of the eclipses change slowly.

The system stays intact for 200 years (energy error 7×10^{-13}). The mutual inclination of the inner and middle orbits stays at 0.38–0.45° (published 0.47 ± 0.25°), that of the outer orbit with the others at 1.0–1.7° (published 5.7°, with an uncertainty of +5.0/−3.9°).

SWINGERsim measures the apsidal period of the 51-d orbit as **32.4 yr** (a 300-yr run: 32.39 yr; the ensemble: 32.2–32.7 yr), and that of the outer orbit as 836–840 yr. The paper quotes 34.79 ± 0.07 yr and 896 yr, but these are, in the words of its Table 1, “the nominal, theoretical apsidal motion periods” from an analytic formula, not measured values. The 7 % difference is therefore between SWINGERsim’s N-body integration and an analytic approximation, not between SWINGERsim and the observations.

4.4 TESS light curves

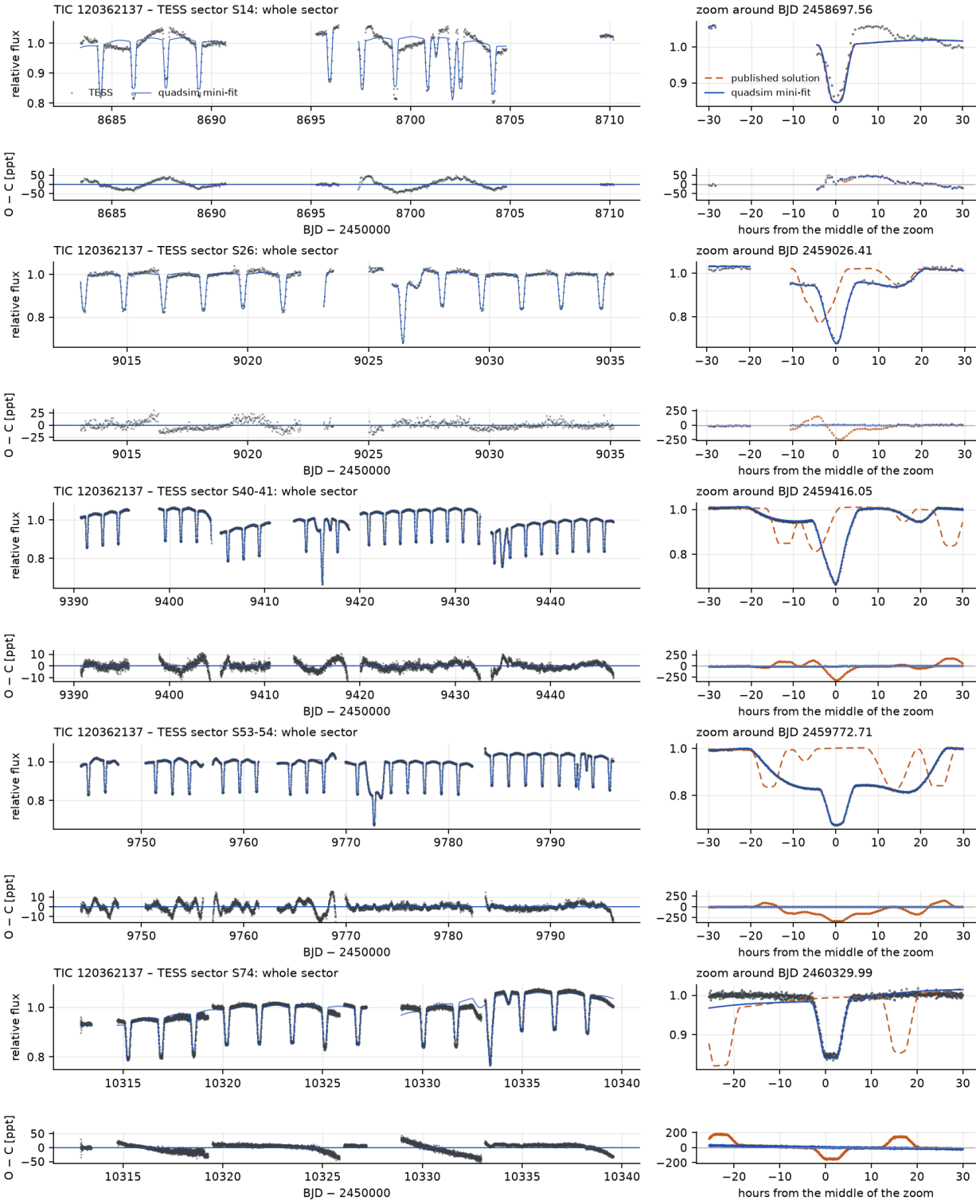


Figure 9. TIC 120362137: TESS light curves of sectors 14, 26, 40–41, 53–54 and 74 with the SWINGERsim model (fitted to the times of minima, blue) and residuals; right: zooms on events with third-body eclipses, with the models of the published elements (orange) and of the fit (blue).

How to read the figure. The regular dips are the eclipses of the 3.3-d pair. The irregular, broader or doubled events (e.g. sector 26 near BJD 2459026.6, sector 53 near 2459772.7) are third-body eclipses: star B passes the inner pair during its 51-d orbit and partly covers one or both of its stars, or is covered by them. These events are extremely sensitive to the

orbits; they are the hardest test of the dynamics.

Table 9. Residuals of the TESS light curves of TIC 120362137 (rms, ppt). Eclipse depths of the inner pair ≈ 150 ppt.

sector	points	published	anchored	fitted to the minima alone
14	778	21.7	21.5	22.2
26	915	32.5	8.3	11.4
40–41	6 763	27.0	3.4	4.1
53–54	6 391	36.1	4.2	6.1
74	9 923	42.9	14.1	15.8

The anchored model reproduces the regular eclipses to 3–4 ppt in sectors 40–41 and 53–54, and the third-body eclipses qualitatively: they appear in the right sectors with the right depths and durations, with their timing off by a few hours. A fit to the minima alone makes the third-body events worse (sector 53–54: 4.2 → 6.1 ppt), because it moves the outer orbit (below). Sectors 14 and 74 have strong instrumental trends in the SPOC photometry that the quadratic baseline cannot remove; their rms is not a property of the model.

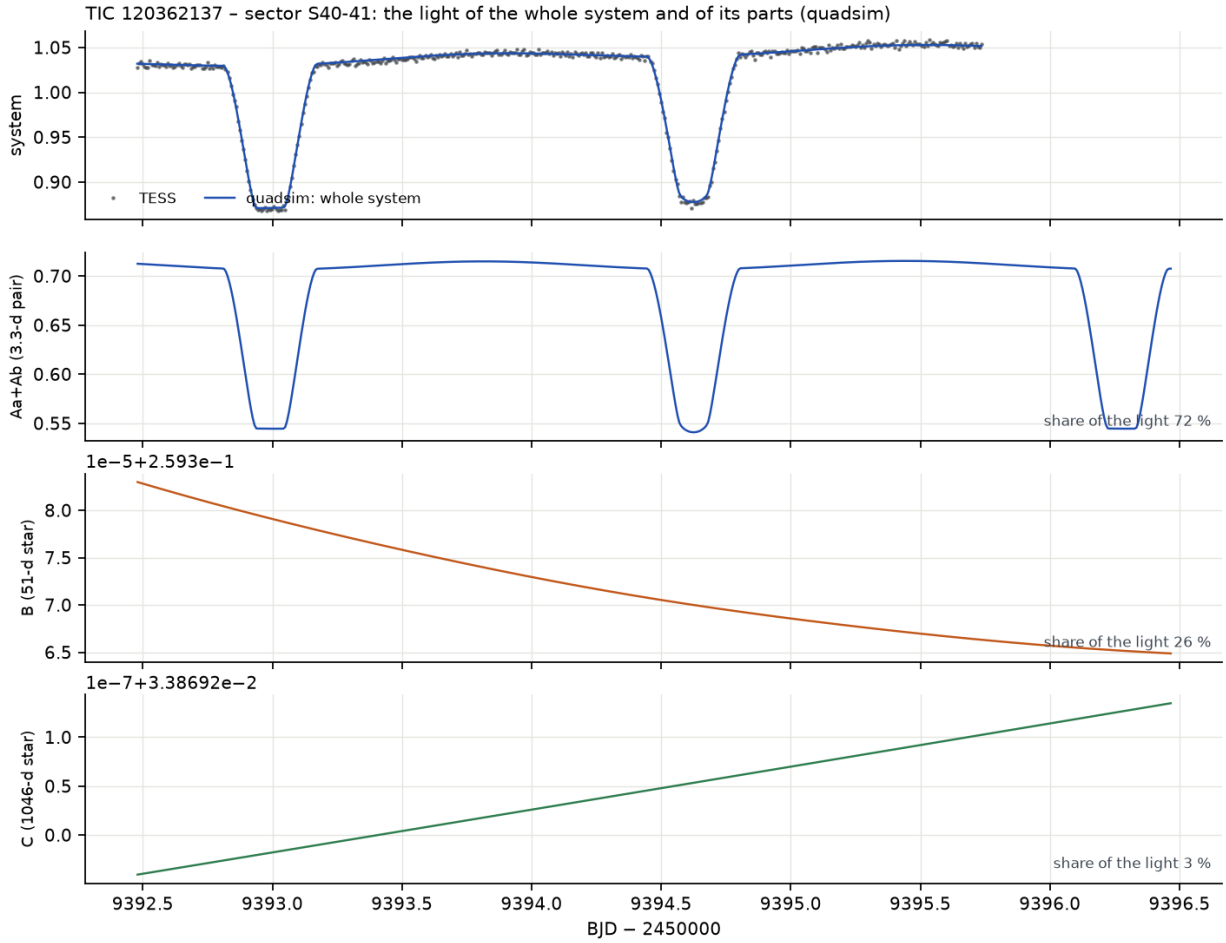


Figure 10. TIC 120362137, sectors 40–41: the light of the whole system and of its parts – the 3.3-d pair (74 % of the light), star B (23 %) and star C (4 %).

How to read the figure. Only the 3.3-d pair varies; B and C contribute a constant light (the tiny drifts in their panels, at the level of 10^{-5}, are the light-travel time and Doppler beaming). During a third-body eclipse the panel of B would dip as well.

4.5 Times of minima

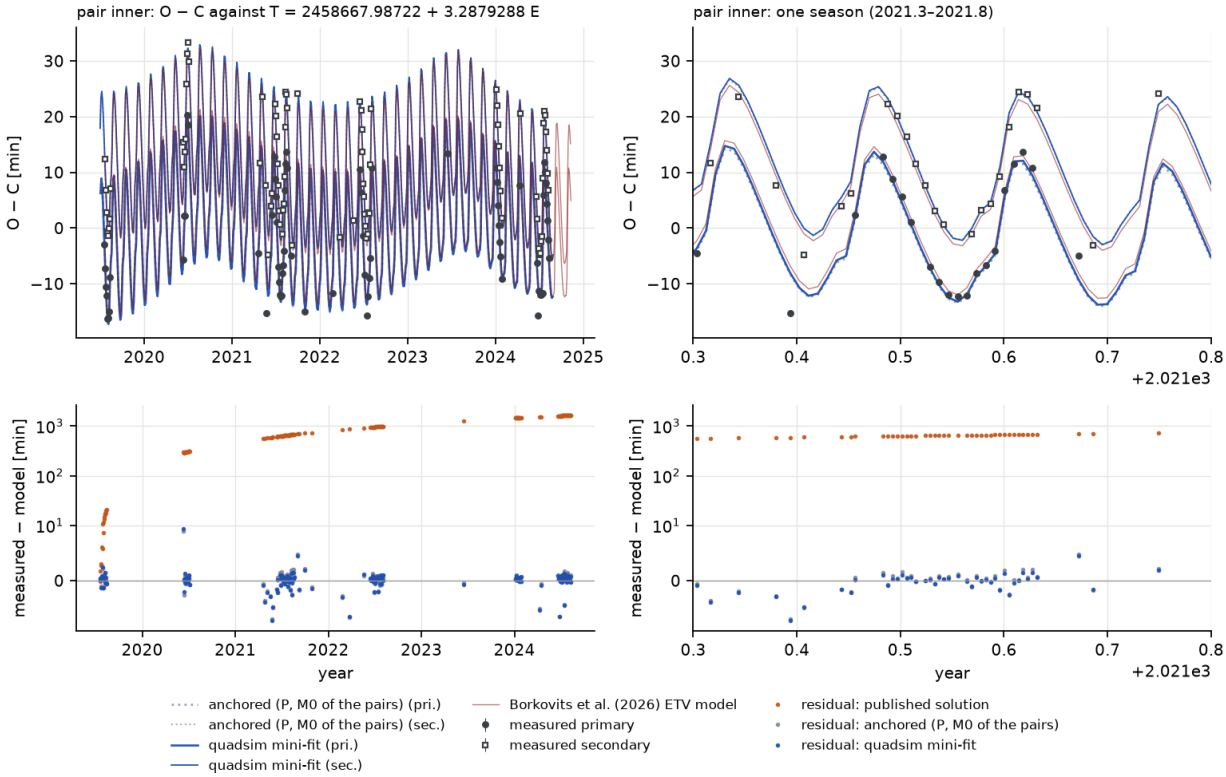


Figure 11. TIC 120362137: O − C of the 3.3-d pair. Top: measured minima and models; bottom: residuals of the three configurations.

How to read the figure. The O − C of the inner pair is dominated by the 51-d orbit of B (dynamical ETV with an amplitude of about ±15 min, every 51 days) and modulated by the 1046-d orbit of C (light time and perturbation). Primary and secondary minima follow slightly different curves because the forced eccentricity of the pair changes.

Table 10. Times of minima of TIC 120362137 (147 minima, 2019–2024).

configuration	rms	median of the absolute O − C
published	1039 min	836 min
anchored	1.74 min	0.59 min
final fit	1.24 min (143 minima)	–
for comparison: the model of Borkovits et al. (2026) on the same minima	1.82 min	–

The published osculating period of the pair corresponds in SWINGERsim to a mean eclipse period shorter by 6.2×10^{-4}; within five years the eclipses drift by 27 hours. After anchoring, SWINGERsim's N-body model reproduces the whole complex ETV, 51-d and 1046-d terms together, to 1.7 min rms, **about as well as the photodynamical model of the paper itself (1.82 min on the same minima, from the model ETV the authors published)**; the final fit reaches 1.24 min. The comparison is not quite like for like: the anchored model tunes two numbers of the pair (period and phase) to exactly these minima, while the published model was fitted to all its data jointly (light curves, velocities, minima and the spectral energy distribution), so a slightly larger rms on the minima alone is expected from it.

4.6 The fit

Table 11. TIC 120362137: what each kind of data adds. Published outer orbit: P_out = 1045.5 ± 7.9 d, e_out = 0.274 ± 0.013, ω_out = 289.0 ± 4.8°.

fit	data	χ^2_red	rms [min]	P_out [d]	e_out	ω_out [°]
minima alone, quoted errors	147 minima	148	1.68	1074 ± 30	0.42 ± 0.13	239 ± 15
+ extra error per source	145	1.00	1.45	1077 ± 23	0.42 ± 0.09	238 ± 27
+ priors	143	1.00	1.15	1044 ± 6	0.278 ± 0.013	289.3 ± 3.3
automatically from the published elements	143	1.00	1.15	1044 ± 6	0.278 ± 0.013	289.4 ± 3.3
+ radial velocities	141 + 380 RV	1.00	1.02	1052.7 ± 3.3	0.277 ± 0.007	275.7 ± 2.0
+ third-body eclipses, red noise, MCMC (final)	143 + 379 RV + 2676 LC	–	1.24	1034.2 ± 2.2	0.271 ± 0.004	284.4 ± 1.6

With the minima alone the outer orbit wanders far from the published one (e_out 0.42, ω_out 239°): with a single eclipsing pair, the outer orbit is seen only through a small modulation of the inner ETV, covered by less than two outer periods, and many combinations of period, eccentricity, ω and phase fit equally well. **With the priors the whole outer orbit returns to the published values within 0.3σ, at a better quality of the timings** (rms 1.15 min). The excursion was the degeneracy of a weakly constrained orbit, not a tension. Radial velocities and the third-body eclipses then fix the outer orbit without needing the priors much (all within 1.4σ of them).

Table 12. TIC 120362137: published values (Borkovits et al. 2026, Tables 1–2) and the final SWINGERsim fit (minima, radial velocities and nine light-curve windows with third-body eclipses from the paper's supplementary data; posterior median and half the 16–84 % range).

quantity	published	SWINGERsim
P_mid [d], e_mid, ω_mid [°]	51.3111 ± 0.0043, 0.22367 ± 0.00072, 190.05 ± 0.32	51.3036 ± 0.0009, 0.2229 ± 0.0002, 190.30 ± 0.08
P_out [d], e_out, ω_out [°]	1045.5 ± 7.9, 0.274 ± 0.013, 289.0 ± 4.8	1034.2 ± 2.2, 0.271 ± 0.004, 284.4 ± 1.6
masses Aa, Ab, B, C [$M_\odot$]	1.748, 1.361, 1.483, 1.008	1.7635 ± 0.0012, 1.3716 ± 0.0023, 1.527 ± 0.008, 0.996 ± 0.014
inclinations inner, middle [°]	88.7, 89.05	88.80 ± 0.02, 88.89 ± 0.02
radii Aa, Ab, B [$R_\odot$]	2.996, 1.507, 1.768	3.001 ± 0.005, 1.568 ± 0.004, 1.921 ± 0.016
radial velocities: rms	–	3.8–4.3 km/s (Aa, Ab, C), 8.0 km/s (B)
third-body eclipses: rms	–	0.7–2.4 ppt in eight windows, 5.9 ppt in one
depth scale a of the windows	ℓx = 0.510 ± 0.007	a = 0.46–0.51

The masses of the inner pair agree with the paper to 1 %, the third star comes out 3 % heavier and the fourth 1 % lighter. The radius of the third star grows by 9 % and that of the secondary of the pair by 4 %; both are constrained mainly by the third-body eclipses, and the third star's radius is the least certain. The depth scale of the windows, a = 0.46–0.51, is the extra light of the neighbouring star that the paper gives (ℓx = 0.510): the supplementary light curves were not corrected for it, unlike the full-sector SPOC light curves of Table 9 (a = 0.97–0.98). Roche surfaces instead of spheres change the residuals by less than 0.1 ppt and the radii by ≤ 2 %: the pair is detached enough for spheres. The mass errors of 0.07–1.4 % are from the MCMC with the correlated noise of the light curves taken into account; without it, the linear errors of a fit to 2676 photometric points were several times smaller and unrealistic.

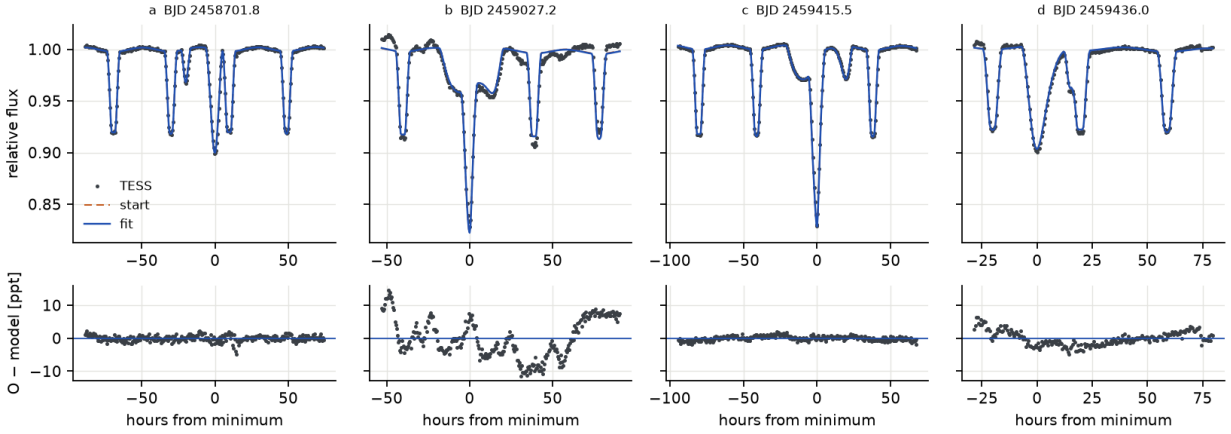


Figure 12. TIC 120362137: four windows of the third-body eclipses (Borkovits et al. 2026, Supplementary Data 1) with the start model and a light-curve fit, and the residuals.

How to read the figure. Each window contains several eclipses of the 3.28-d pair and one eclipse involving the third star (the deep or wide extra dip). The fit reproduces the third-body eclipses to about 1 ppt in most windows; window b (BJD 2459027) keeps a structured residual of ±10 ppt.

Predictions. The final fit predicts the next third-body events about every 25 days through 2027 (e.g. BJD 2460559, 2460587, 2460610, 2460638, … 2461480). Testing them needs new TESS sectors or ground photometry.

4.7 Ensemble

400 systems within the published errors (masses, the middle and outer orbits, nodes) were integrated 100 years forward and backward. **All 400 stay intact and stable**, and the inner pair eclipses in all of them. The apsidal period of the 51-d orbit is 32.4 yr (16–84 %: 32.2–32.7 yr) in all of them.

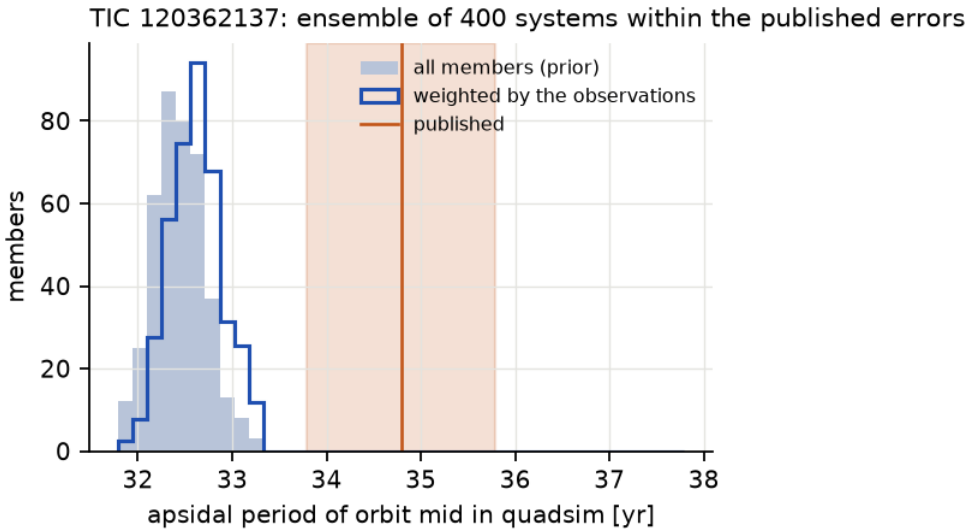


Figure 13. TIC 120362137: distribution of the apsidal period of the 51-d orbit in the ensemble, with the published (theoretical) value.

How to read the figure. As for BU CMi. The narrow distribution shows that the N-body apsidal period hardly depends on the uncertainties of the masses and orbits; the published value is the analytic estimate of the paper.

4.8 Verdict

- **Confirms** the published solution: stable (stability ratios 1.5–1.9), the complex ETV reproduced to 1.7 min after anchoring the period and phase to the minima (about as well as the jointly fitted published model), the TESS light curves to 3–4 ppt with the third-body eclipses in the right places, and, in the joint fit, the masses to 1–3 %, the inclinations to 0.15° and the third-body eclipses to 1–2 ppt.
- **Adds:** the N-body apsidal period of the 51-d orbit, 32.4 yr (the published 34.79 yr is a theoretical value), and predictions of the next third-body events.
- **Shows a limit of timing-only fits:** with one eclipsing pair the outer orbit is not determined by the minima alone; priors, velocities or third-body eclipses are needed.

5 V994 Her ((2+2)+2)

5.1 The system according to Zasche et al. (2023)

V994 Her contains six stars in three eclipsing binaries: A (2.08 d) and B (1.42 d) orbit each other in 1062 days on a very eccentric orbit (e = 0.69), and the third binary C (1.96 d) is a visual companion 1.1″ away, at a projected distance of about 300 au. The stars are young (about 80 Myr) B and A stars of 1.1–2.9 solar masses; all three pairs have slightly eccentric orbits and measurable apsidal motion. The solution of Zasche et al. (2023, Table 3) combines TESS light curves, radial velocities of A and B (Lee et al. 2008), the ETVs of all three pairs and the spectral energy distribution. The minima span twenty years, and many come from observers: S. Dallaporta and U. Munari (SD & UM, 2002–2004), R. Uhlař (RU, 2017–2022, 64 minima from a private observatory), the robotic telescope FRAM, M. Mašek (MM) and others (Zasche et al. 2023, Tables A1–A4).

The papers give neither the orbital nodes nor the orbit of C around A+B; we assumed Ω = 0 for all orbits and an orbit of C with P = 3000 yr, e = 0.3. The orbit of C is so wide that it does not affect the timings over 20 years except through a nearly constant light-time term.

5.2 Stability

Table 13. Stability of V994 Her (with the assumed orbit of C).

sub-triple	a_in / a_out	e_in	e_out	i_mut	MA01	V22
A inside A–B (+B)	0.013	0.028	0.687	1.7°	4.43	6.26
B inside A–B (+A)	0.009	0.119	0.687	6.2°	5.65	7.53
A–B inside the orbit of C (+C)	0.009	0.687	0.300	0.0°	21.0	19.9

All sub-systems are stable by a wide margin, and the classifier of Vynatheya et al. (2023) gives *stable* for each sub-quadruple. The ghost test over one orbit of C (3000 years, about 1000 orbits of A–B) finds the system stable (divergence 2.3×10^{-5}).

5.3 Thirty years forward and backward, and the apsidal motion

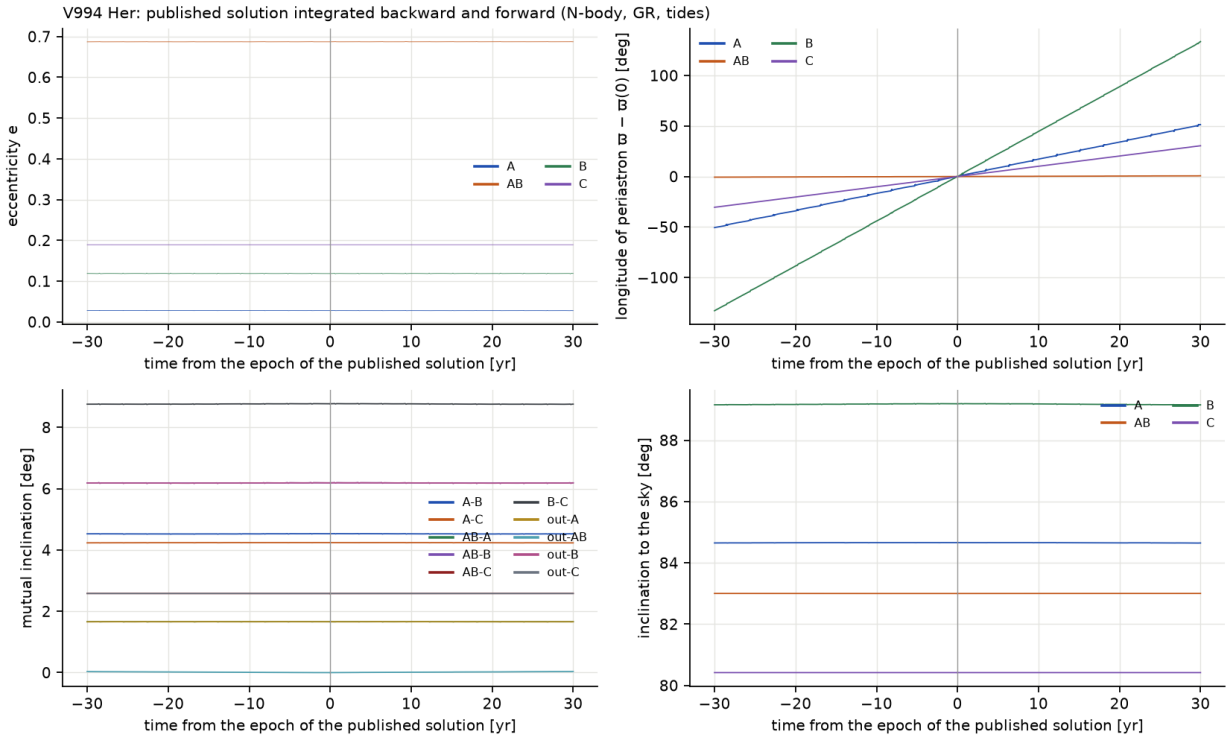


Figure 14. V994 Her: the published solution integrated 30 years backward and forward; panels as for BU CMi (orbits A, B, C, A–B and the assumed orbit of C, "out").

How to read the figure. As for BU CMi. Over 60 years almost nothing changes: the orbits are much wider relative to the pairs than in BU CMi, the perturbations are small, and the slow apsidal motion of the pairs (periods of a century and more) shows up as straight lines in the top right panel.

The system stays intact (energy error 5.6×10^{-11}), the eccentricities stay within 0.0274–0.0279 (A), 0.118–0.120 (B) and 0.6869–0.6874 (A–B), and all three pairs eclipse throughout. Unlike in BU CMi, the precession is mostly *classical*: caused by the tidal and rotational bulges of the stars themselves, not by the other pair.

Table 14. Apsidal periods of V994 Her. Published: 360° / ω̇ from Table 3 of Zasche et al. (2023).

pair	published	SWINGERsim run	SWINGERsim ensemble, weighted (16–84 %)	pair alone (tides + rotation + GR)
A	194 (158–271) yr	212 yr	213 (204–222) yr	244 yr
B	98 (94–103) yr	81 yr	81 (80–83) yr	84 yr
C	214 (188–250) yr	355 yr	355 yr	355 yr

For A SWINGERsim agrees within the (large) published uncertainty. For B SWINGERsim's precession is 17 % faster, for C 40 % slower (a period 66 % longer). The two models do different things here. Zasche et al. (2023) fitted a constant apsidal rate $\dot{\omega}$ for each of the three pairs as a free parameter of their analytic Keplerian model; SWINGERsim computes the rates from the stars: the apsidal constants k_2 (Claret 2023, at the MIST age), the radii and the rotation. The differences therefore compare a measured rate with the rate that stellar structure predicts. For C SWINGERsim's prediction is 0.78°/yr tidal, 0.10°/yr rotational and 0.14°/yr relativistic; pseudo-synchronous instead of synchronous rotation adds only 0.05°/yr. Reaching the published 1.68 ± 0.24°/yr would need k_2 of both stars 1.76 times larger, far beyond the uncertainty of stellar models (about 10 %). B would need k_2 multiplied by 0.81, A (with the perturbation by B, +0.22°/yr) by 1.12.

For B, Zasche et al. (2023, Sect. 4.1) made the same comparison themselves. After subtracting the relativistic and dynamical parts they derive from the fitted rate $\log k_2 = -2.44 \pm 0.05$, against -2.36 ± 0.02 from the models of Claret (2004), i.e. a measured k_2 about 0.83 times the theoretical one. They discuss a slightly faster apsidal motion, sub-synchronous rotation or a higher metallicity as explanations and find that metal-rich models ($Z = 0.04$, $\log k_2 = -2.40$) agree better. Our factor of 0.81 for B therefore reproduces the authors' own finding with a different set of stellar models; it is a confirmation, not a new discrepancy. They do not make the comparison for A and C. C is not perturbed in SWINGERsim (its outer orbit is assumed wide); the published rate of C is 2.8σ from the structural prediction, and a closer, unknown companion of C would be one explanation.

A hidden companion of C. To see which companion would do it, 3000 configurations of C with a third star were integrated for 30 yr (0.1–2 $M_\odot$, period 8–3000 d log-uniform, eccentricity 0–0.6, isotropic orientation, MA01-stable): 2592 stay intact, 403 collide through ZLK cycles (mutual inclinations 72–135°, periods ≤ 152 d) and 5 are ejected. Without a companion (periods above 2000 d) C precesses in 354 yr, as above. 172 members reach the published 214 ± 31 yr. Weighted by it, the companion has 0.17–0.44–1.36 $M_\odot$ on a 92–171–332-d orbit (16–50–84 %), which shifts the eclipses of C by 0.7–8.5 min through light-time and by 0.4–1.5 min dynamically. The 56 primary minima of C (2003–2022, mostly TESS, rms 3.8 min around a cubic ephemeris) limit any periodic signal of 80–400 d to below 2.4 min (2σ, median over the period). That excludes 55 % of these companions. The remaining ones have 0.13–0.20–0.31 $M_\odot$ on 80–126–182-d orbits. The highest peak in the minima (80.9 d, 2.0 ± 0.5 min) is not a detection, because aliases at 49–84 d are almost as high. A low-mass companion of C on an orbit of a few months therefore remains possible. Dense timing of C over at least two such periods would find or exclude it.

5.4 TESS light curves

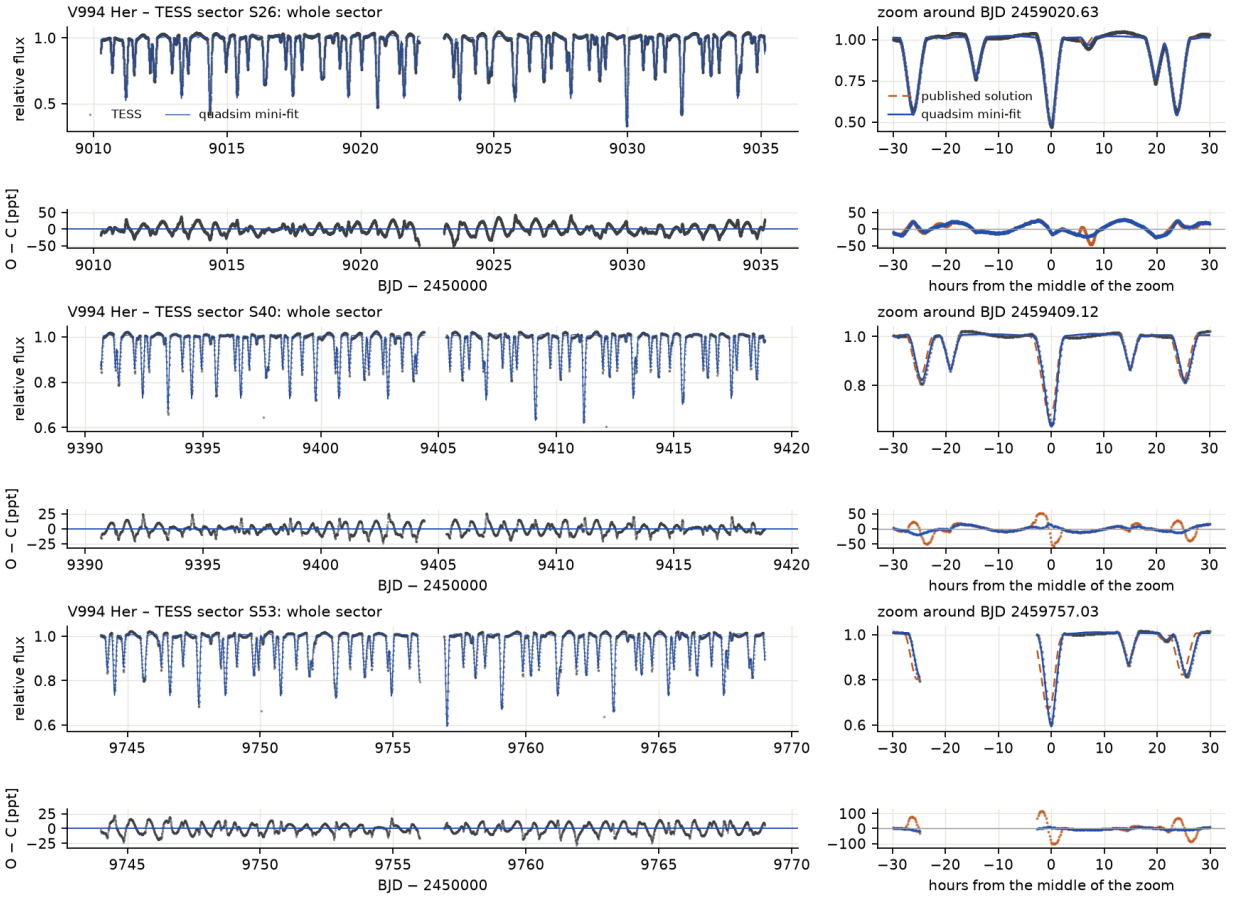


Figure 15. V994 Her: TESS light curves of sectors 26 (SPOC, 2 min), 40 and 53 (QLP, 10 min) with the SWINGERsim model (blue), residuals, and zooms with the models of the published elements (orange) and of the fit (blue).

How to read the figure. Three binaries eclipse in the same light curve, so the dips come in three interleaved rhythms (2.08, 1.42 and 1.96 days) and different depths; the deepest dips (to 0.5 of the flux in sector 26) are eclipses of the brightest star of A. The zooms show that the published elements already place the eclipses well (V994 Her is less compact, and its osculating and mean periods differ less) and that the fit moves them onto the data.

Table 15. Residuals of the TESS light curves of V994 Her (rms, ppt). Eclipse depths 50–550 ppt.

sector (cadence)	points	published	anchored	fitted to the minima	scale a of the depths
26 (SPOC, 2 min)	16 931	15.6	14.9	14.5	2.03
40 (QLP, 10 min)	3 917	17.1	8.8	8.0	1.20
53 (QLP, 10 min)	3 454	18.2	9.4	8.2	1.18

Two things stand out.

An oscillation. The residuals of 8–15 ppt are larger than in the other systems and show a regular oscillation with a period of 0.663 d, present in QLP sectors 40 and 53 and in SPOC sector 26, with an amplitude of 7.5–9 ppt in QLP and 15 ppt in SPOC (the same factor as the eclipse depths, see below). Zasche et al. (2023, Sect. 3) found it before us: an oscillation

of about 2 per cent with the period P_A − P_B (2.0833 − 1.4200 d = 0.663 d), which they removed before fitting; it is not part of either model. When it is removed from the light curve (a sum of sine waves at its period and harmonics, fitted outside the eclipses), the residuals in sectors 40 and 53 fall to 3.4–3.6 ppt.

The depths of sector 26. The depths must be scaled by 1.2 in the QLP sectors and by 2.0 in the SPOC sector 26. A scale above 1 means that the observed eclipses are deeper than SWINGERsim predicts for the total light of the six stars, i.e. the photometry contains less light than the system emits. In sector 26 the SPOC eclipses are 1.75 times deeper than in QLP (primary of A: 0.44 against 0.25). A likely cause is the crowding correction of the SPOC pipeline (CROWDSAP), which counts the light of the 1.1″ companion C and of neighbouring stars as contamination and subtracts it; we have not proved it. Zasche et al. (2023, Sect. 3) also needed a separate extra light for sector 26 (0.058, against 0.181 for sectors 40 and 53), but their case is not the same as ours: for sectors 40 and 53 they used their own photometry of the full-frame images (the FITSH package), not QLP, and their sector-26 eclipses were only "a few per cent" deeper than in the other two sectors. The large factor we find therefore depends on the pipelines we used (SPOC against QLP) and tells nothing about the stars.

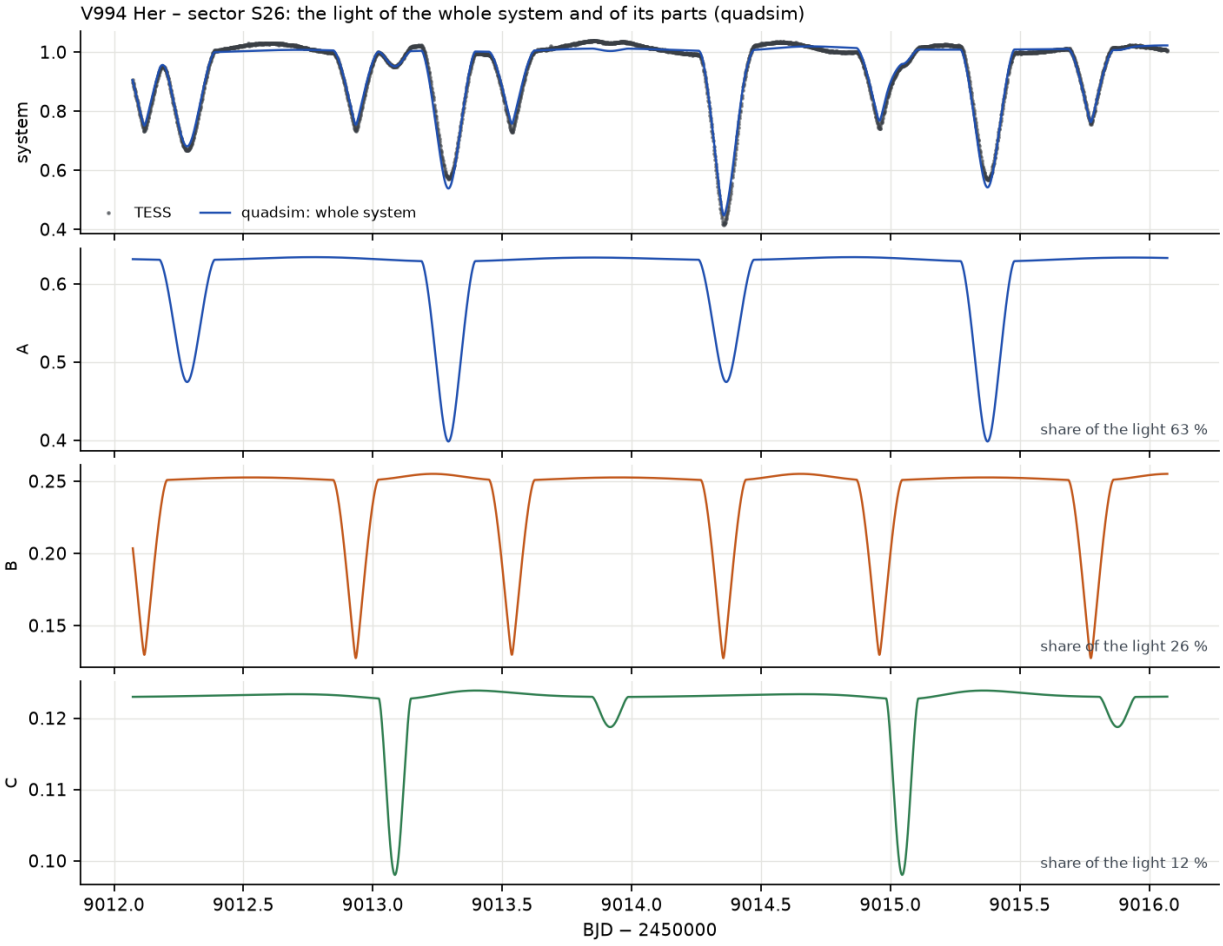


Figure 16. V994 Her, sector 26: the light of the whole system and of the three pairs A (63 % of the light), B (25 %) and C (12 %).

How to read the figure. The three lower panels separate what TESS records together. C is the faintest pair and its eclipses are shallow in the total light, which is why its minima are the least precise.

5.5 Times of minima

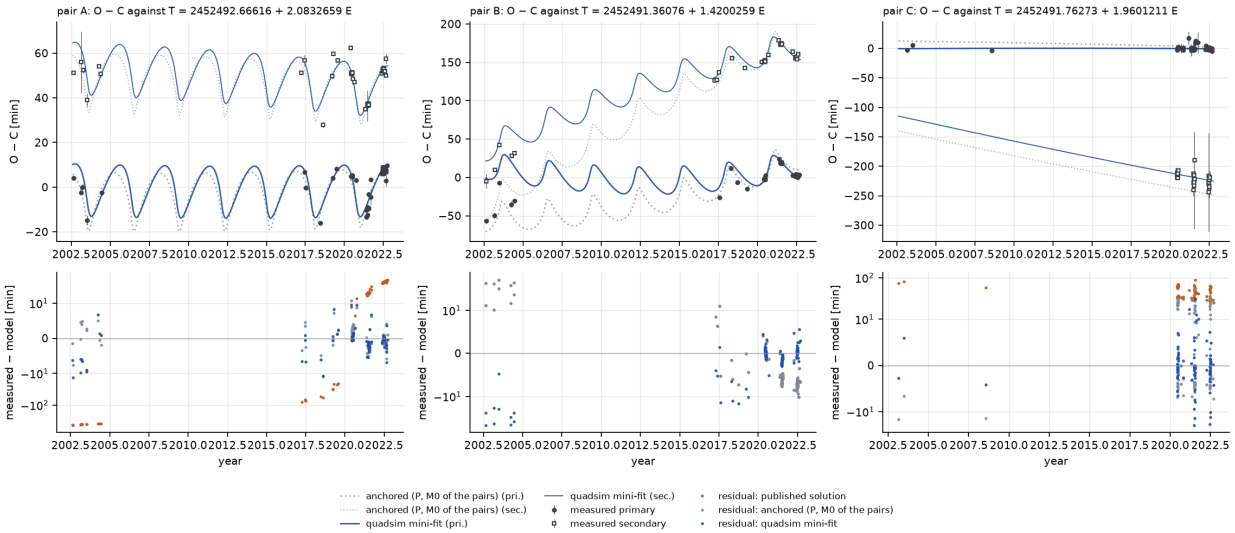


Figure 17. V994 Her: O − C of pairs A, B and C (top) and residuals of the three configurations (bottom).

How to read the figure. For A and B the waves with a period of 2.9 years are the light-travel time across the eccentric A–B orbit: when A is on the far side, its eclipses come later and those of B earlier, so the waves of A and B are mirror images with amplitudes in the ratio of the masses. The slower opposite drifts of primary and secondary minima are the apsidal motion. For C the primary and secondary minima lie 200 minutes apart because of its eccentric orbit (e cos ω ≠ 0); the slow divergence is its apsidal motion.

Table 16. The minima of V994 Her by source, and the extra error that the fit assigns to each source.

source	years	minima	median quoted σ	extra error found by the fit
SD & UM (Dallaporta, Munari)	2002–2004	23	128 s	10 min
RU (R. Uhlař)	2017–2022	64	60 s	4.9 min
TESS	2020–2022	246	10 s	1.1 min
FRAM	2021–2022	7	45 s	2.2 min
other observers (MR13, GP, MM)	2008–2021	3	26–86 s	5.5 min

The recent ground-based minima of the observers (RU, FRAM, MM) agree with the model nearly as well as TESS; the 2002–2004 minima lie on average 17 minutes earlier than the model and need an extra error of 10 min. They are the only minima of the first decade and anchor the long-term trends. The published elements give errors of hours only for A; after anchoring, the model reproduces A to 3.3 min and the LTTE waves of A and B together, which confirms the masses and the A–B orbit.

5.6 The fit

Table 17. V994 Her: what each kind of data adds. Published A–B orbit: P_AB = 1062.3 ± 2.6 d, e_AB = 0.687 ± 0.044, ω_AB = 59.6 ± 3.8°.

fit	data	χ^2_red	rms A / B / C [min]	e_AB	ω_AB [°]
minima alone, quoted errors	338 minima	86	3.8 / 10.4 / 6.3	0.579 ± 0.016	30 ± 5
+ extra error per source	337	1.00	4.0 / 3.5 / 5.3	0.511 ± 0.013	21 ± 8
+ priors	337	1.00	5.0 / 3.4 / 5.4	0.570 ± 0.018	51 ± 3
automatically from the published elements	337	1.00	5.1 / 3.8 / 5.4	0.569 ± 0.018	51 ± 3
+ radial velocities	337 + 128 RV	1.00	7.4 / 5.2 / 5.3	0.590 ± 0.018	56 ± 3
+ light curves with the oscillation removed, red noise (final)	333 + 128 RV + 9214 LC	–	5.3 / 3.4 / 5.4	0.587 ± 0.018	52 ± 3
radial velocities, no SD & UM minima, no priors	314 + 128 RV	1.00	3.0 / 2.2 / 5.3	0.533 ± 0.026	40 ± 8

The extra error per observer is the most effective single step: the rms of pair B falls from 10.4 to 3.5 min, because the oldest minima no longer dominate. The eccentricities and arguments of periastron of the three pairs agree with the published ones within 2σ in every variant, a good independent confirmation. The A–B orbit, however, is seen in the minima only through the light-time waves, and the fit prefers e_AB = 0.51–0.59 instead of 0.69 ± 0.04. With priors it stays 2.3σ below, and dropping the early SD & UM minima does not bring it back (0.53). The published value rests also on the radial velocities of the A–B orbit and on the light curves in a model that includes more than the timings; the difference remains open.

The radial velocities and the masses of pair A. The velocities of Lee et al. (2008), 32 per star from quadruple-lined spectra taken in March–May 2006, pull the masses of pair A in the joint fit down to 2.38 ± 0.07 and 1.87 ± 0.06 $M_\odot$ (published 2.93 and 2.22 $M_\odot$) and leave pair B unchanged (1.92 and 1.97 against 1.91 and 1.89 $M_\odot$). A simple Keplerian fit of the same velocities, with a linear trend of the systemic velocity of each pair over Lee's 60 days, gives K_A = 126–128 and 157–159 km/s, i.e. 2.82 and 2.29 $M_\odot$ at i = 84.7°, in agreement with Zasche et al. (2.93 and 2.22 $M_\odot$). Over Lee's 60 days the systemic velocity of A − B changes by −11.8 km/s per 30 days; the published A–B orbit gives −12.2, the joint fit −9.1. The low masses of the joint fit are therefore an artefact: with its less eccentric A–B orbit, the fit absorbs part of the orbital motion of the pairs into their velocity amplitudes. Lee et al. (2008) quote neither errors of the individual velocities nor the rms of their orbital fits; they only note systematic offsets in pair B, which they ascribe to the blending of the lines of four stars or to an unresolved companion. The errors of 5 km/s per velocity used here were therefore assigned in this test, and the fit's extra error per source decides how much the velocities are trusted.

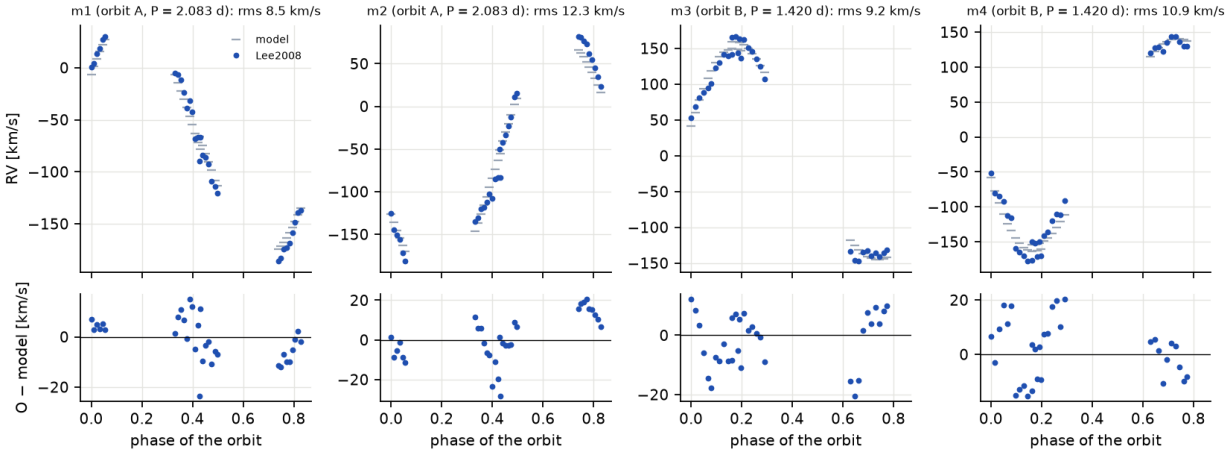


Figure 18. V994 Her: radial velocities of the four stars of A and B against the phase of their pair (Lee et al. 2008), with the model and the residuals.

How to read the figure. As for BU CMi. The model follows the velocity curves of all four stars; the residuals of 9–11 km/s are about twice the errors of 5 km/s that we assigned to the velocities of Lee et al. (2008), who give no errors themselves.

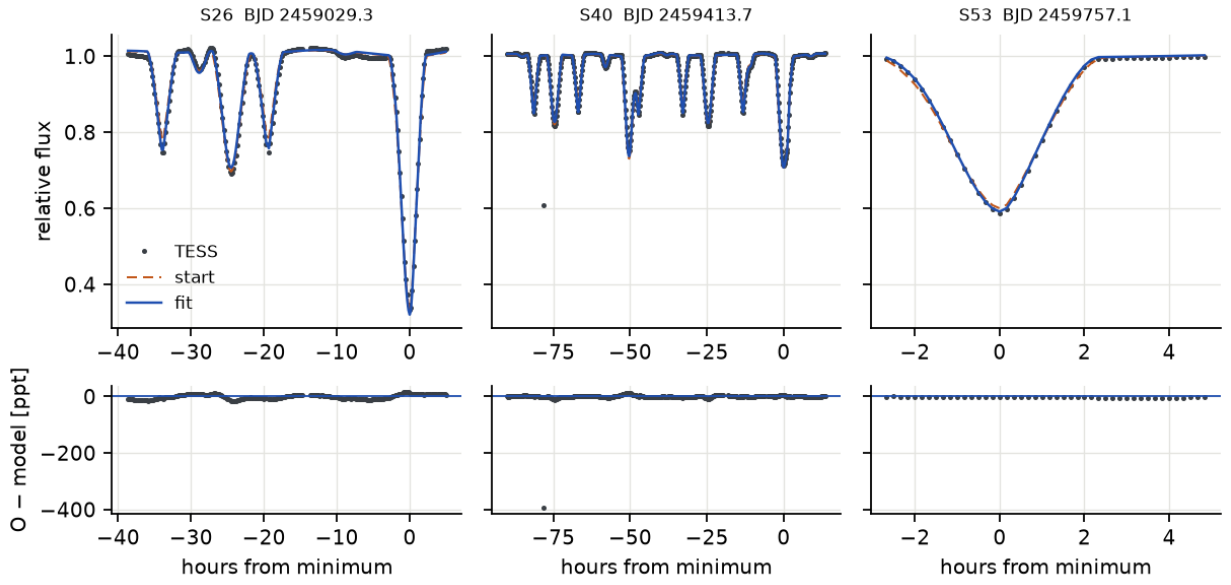


Figure 19. V994 Her: four stretches of the TESS light curve (oscillation removed) with the start model and the final light-curve fit, and the residuals.

How to read the figure. The windows around the eclipses of the three pairs overlap, so that each panel shows one to four days. With the 0.663-d oscillation removed from the data, the residuals are a few ppt (with it, they were ±20 ppt).

With the oscillation removed, the light-curve fit of sectors 26, 40 and 53 gives 3.4–3.6 ppt in the QLP sectors (9.7 ppt in sector 26), inclinations of 85.4 ± 0.8° (A), 89.0 ± 0.8° (B) and 80.6 ± 0.3° (C, published 84.66°, 89.19°, 80.42°), and radii within 5 % of the published ones except the secondary of A (1.49 against 1.72 $R_\odot$, the radius-ratio degeneracy of partial eclipses).

5.7 Ensemble

500 systems were drawn with the published masses and A–B orbit within their errors, random nodes of A and B within ±30°, a random node of C, and a random orbit of C around A+B

(period 500–20 000 yr, eccentricity 0–0.7, inclination 40–140°), keeping only those stable by MA01. Integrated 30 years forward and backward, **all 500 stay intact**, and all three pairs keep eclipsing. Weighted by the published apsidal periods and the eclipses, n_eff = 243.

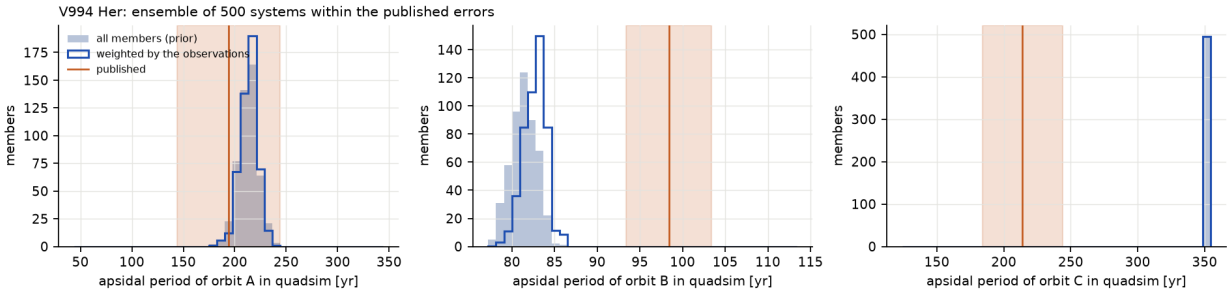


Figure 20. V994 Her: distributions of the apsidal periods of A, B and C in the ensemble, with the published values and ranges.

How to read the figure. As for BU CMi. For A the ensemble covers the published value. For B the whole ensemble lies at 78–86 years, well below the published 98 years: no system within the published masses and orbits reaches the published precession. For C all members have the same 355 years: its precession is entirely classical, and the random orbit of C around A+B never changes it. Both differences are therefore between the measured (fitted) rates and the stellar structure, not properties of the orbits.

5.8 Verdict

- **Confirms** that SWINGERsim handles a six-star, three-level hierarchy with three eclipsing pairs: stable, the light-time waves of A and B reproduced from the published masses, the eccentricities and orientations of the three pairs within 2σ of the published ones, the eclipse timings of all three pairs to 3–5 min (TESS to 1 min), and the light curves to 3.5 ppt once the oscillation is removed.
- **Confirms** the oscillation with the period P_A − P_B = 0.663 d that Zasche et al. (2023) removed, and the authors' own finding that the apsidal motion of B corresponds to an apsidal constant about 0.8 times the theoretical one (their 0.83, our 0.81).
- **Adds:** realistic errors for every observer (TESS 1.1, FRAM 2.2, R. Uhlař 4.9, SD & UM 10 min).
- **Differs, explained:** the classical apsidal motion of C, which SWINGERsim predicts from stellar structure while the paper fitted it freely (C would need k_2 1.8 times larger than stellar models give, or a companion of 0.13–0.31 $M_\odot$ on an 80–182-d orbit, which the present minima of C still allow); the low masses of pair A in the joint fit, an artefact of the less eccentric A–B orbit (Lee's velocities alone agree with the published masses).
- **Open:** the A–B eccentricity, which the minima put at 0.51–0.59 instead of 0.69 ± 0.04; the depth scale of 2.0 in SPOC sector 26 against 1.2 in QLP, probably the SPOC crowding correction but not proved, and not directly comparable with the paper, which used different photometry for sectors 40 and 53.

6 The three systems compared

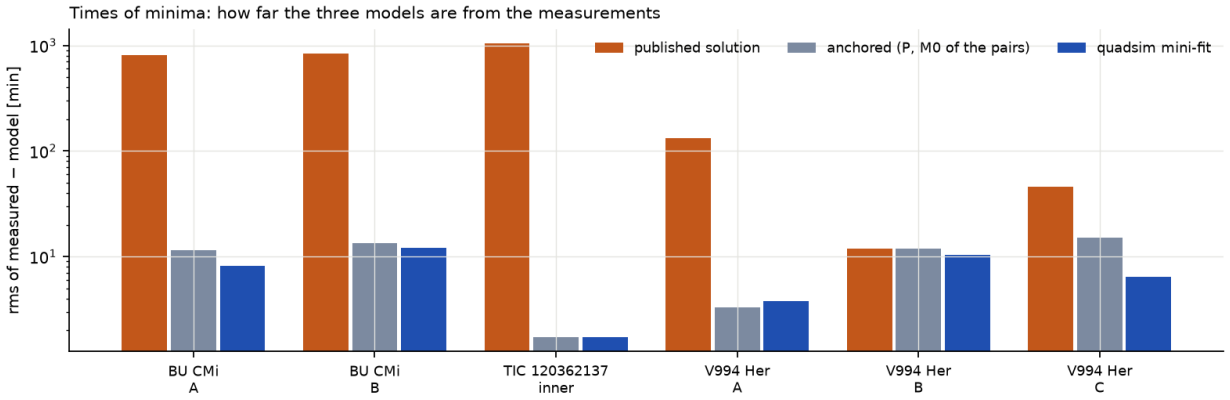


Figure 21. Times of minima: rms of measured − model for the three configurations of every system and pair (logarithmic scale).

How to read the figure. Each group of three bars is one eclipsing pair. Orange: the published elements taken literally; grey: after anchoring the period and phase of each pair; blue: after SWINGERsim's fit to the minima. The logarithmic axis spans three orders of magnitude: from days of drift to a few minutes.

Table 18. Summary of the test. *yes*: agreement; *partly*: agreement with a qualification; *no*: a difference, explained in the text.

question	BU CMi	TIC 120362137	V994 Her
stable (MA01, V22, ML, ghost test)	yes	yes (ratios 1.5–1.9)	yes
intact in all ensemble members, forward and backward	yes – 600/600	yes – 400/400	yes – 500/500
dynamical apsidal motion vs published	yes – 1 % (point masses)	– (published value theoretical)	– (mostly classical)
classical apsidal motion	adds 2–4 %; the tidal term of the published integration is not stated	–	A yes, B −17 % (as the paper's own k_2 for B), C +66 % (fitted freely in the paper)
TESS light curves (best sectors, anchored)	yes – 2–3 ppt	yes – 3–4 ppt	yes – 3.5 ppt after removing the oscillation
third-body eclipses	–	yes – 1–2 ppt (fit)	–
times of minima, anchored	yes – 11–13 min (survey minima)	yes – 1.7 min	yes – 3.3 / 11.9 / 15.2 min
times of minima, final fit	7.6 / 9.1 min; TESS and follow-up ~3 min	1.2 min	5.3 / 3.4 / 5.4 min; TESS ~1 min
masses (joint fit) vs published	yes – 1–5 %	yes – 1–3 %	partly – B yes, A −19 % (artefact, see text)
inclinations vs published	yes – 0.1°	yes – 0.15°	yes – within 1°

question	BU CMi	TIC 120362137	V994 Her
outer orbit vs published	partly – e_out −2.6σ with tides, agrees without	yes – within 1.4σ	partly – e_AB −2.3σ
radial velocities	no – rms 17–25 km/s (quoted 2–3)	yes – 3.8–4.3 km/s (B: 8.0)	partly – 9–11 km/s (no published errors; 5 km/s assumed)

7 Discussion

7.1 Published elements are not portable without anchoring

The most visible result is that the published elements, entered literally into SWINGERsim, produce eclipses that are off by hours to days after a few years. The orbits are not wrong: the same osculating numbers mean slightly different motions in two codes that use different coordinates for the hierarchy and different models for the tidal and rotational forces of the stars. A relative difference of a few 10^{-4} in the mean period is invisible in the light curve of one TESS sector but adds up to hours over thousands of cycles. Two numbers per pair (period and phase) remove the effect completely. Anyone who transfers a photodynamical solution into an N-body code should expect this and anchor the solution to the minima first; SWINGERsim does it automatically when a fit starts from published elements.

7.2 The dynamics is right; the classical apsidal motion tests stellar structure

Where the apsidal motion is driven by the other stars, SWINGERsim reproduces the published integration: BU CMi to 1 % with point masses, and the full ETV of TIC 120362137 as well as the authors' own model. Where the stars themselves contribute, the models differ in what they assume. In BU CMi SWINGERsim adds the classical advance that stellar models predict (2–4 % of the total), and the fit compensates it through the outer eccentricity (2.6σ). The published integration behaves like SWINGERsim's point-mass model, although the paper's own analytic estimate of the tidal advance (0.65 and 0.39°/yr) is as large as SWINGERsim's (0.56 and 0.33°/yr) and the published code can include tides and rotation; whether its tidal term was smaller or switched off is not stated, and we leave it open. With the apsidal constants free, the minima want an even weaker classical advance ($k_2 \leq 0.0023$ for pair A at $\Delta\chi^2 = 4$ against 0.0042 of the models) but leave the outer eccentricity at 0.266, so the constants do not explain the difference. In V994 Her the paper fitted the apsidal rates freely, while SWINGERsim predicts them; B and A would need apsidal constants changed by about 20 % and 10 %, plausible for stellar models, C by a factor of 1.76, which no model gives. A companion of C of 0.13–0.31 $M_\odot$ on an 80–182-d orbit would do it instead; the present minima of C neither show nor exclude it. For B, Zasche et al. (2023) found the same themselves: their measured k_2 is 0.83 times the theoretical value, which they relate to a possibly higher metallicity; SWINGERsim's factor is 0.81. In both systems, therefore, the differences are not errors of either code but a measurement of the classical apsidal motion: the timings of the next decade will tell how large it is.

7.3 What the fit can and cannot do

- **Extra errors per source** are the single most effective element of the fit. They bring the reduced χ^2 to 1 in all three systems, measure how much each observer's minima can be trusted (from 1 min for TESS to 10–18 min for surveys and early ground minima) and stop a few old minima from pulling the solution.
- **Times of minima alone** determine the eclipsing pairs well (V994 Her: within 2σ of the published values), but not orbits seen only indirectly: the outer orbit of TIC 120362137, seen through one eclipsing pair, and the A–B orbit of V994 Her, seen through light-time

waves. There the fit wanders to solutions that fit the timings equally well but contradict other data.
- **Priors** do what they should for an orbit that the minima do not determine (TIC 120362137), and they reveal the orbits that the minima do determine differently from the papers (BU CMi, V994 Her).
- **Radial velocities** give masses, but they test the velocities as much as the model: in TIC 120362137 they agree to their errors, in BU CMi they scatter far beyond their quoted errors, in V994 Her they agree with the published masses on their own but are partly absorbed by the A–B orbit in the joint fit.
- **Light curves** fix inclinations and radii where the eclipses are clean, leave the ratio of the radii of partially eclipsing stars undetermined without a spectroscopic light ratio, and need any additional signal (the oscillation of V994 Her) removed first. Their correlated noise must be taken into account; without it the errors were several times too small.
- **MCMC with 1000 steps** gives errors that agree with the linear ones to about 10 % (median), and changed the errors of a 400-step run by a few percent. The chains cover about 19 autocorrelation times in total and about ten after the burn-in half is dropped, enough for the quoted ranges but not a formal proof of convergence.

7.4 The role of the observers

The TESS minima cover at most five years. The apsidal motion of BU CMi (25–29 yr) and of V994 Her (a century and more), the 2.9-year A–B orbit of V994 Her and the outer orbit of TIC 120362137 are visible only thanks to ground-based minima that extend the baseline to 15–20 years. In V994 Her the 64 minima of R. Uhlař and those of FRAM agree with the model to 2–5 minutes, close to TESS. The oldest minima (2002–2004 in V994 Her, the surveys of 2008–2016 in BU CMi) are the least precise and the most decisive for the long-term terms. Light curves of the observers were not available to us, only the times of minima derived from them, so the light-curve comparison uses TESS only; ground-based light curves, e.g. from the SQUADRA observers processed with the VYVAR pipeline, would allow the same comparison for the eclipses between the TESS sectors.

7.5 Limitations of this test

- Temperatures were taken from the papers; for V994 Her the nodes and the orbit of C are assumptions.
- SWINGERsim's light curves use limb-darkened spheres or Roche surfaces; the pairs in these systems are detached (Roche lobes filled to ≤ 57 %), and Roche surfaces changed the results by less than the errors.
- The oscillation of V994 Her was removed as a sum of sine waves, not modelled.

8 Conclusions

1. **SWINGERsim reproduces the three published systems.** All are stable in every test, and all 400–600 members of each ensemble stay intact forward and backward in time.
2. **Light curves.** After the period and phase of each pair are anchored to the minima, the synthetic TESS light curves agree with the observations to 2–4 ppt in the best sectors, about one per cent of the eclipse depths (V994 Her after removing a 0.663-d oscillation). The depths are those of the published stars multiplied by one free scale factor per sector, which absorbs extra light in the aperture: 0.98 for BU CMi and 0.97–0.98 for the SPOC sectors of TIC 120362137, but 0.46–0.51 for the uncorrected light-curve windows of TIC 120362137 (a bright neighbour) and 1.2–2.0 for V994 Her.
3. **Eclipse timings.** SWINGERsim reproduces the complex O − C diagrams (apsidal motion, light-travel time, dynamical ETVs) of all six eclipsing pairs, to 1.2–1.7 min for TIC 120362137, as well as or better than the published model on the same minima, and to the scatter of the data elsewhere.
4. **Joint fit.** With an extra error for every source of minima, priors where the data are weak, radial velocities, light curves with correlated noise, and MCMC, the fit reproduces the published masses to 1–5 % and inclinations to 0.15° in BU CMi and TIC 120362137.

5. **Three differences, traced as far as the papers allow.** BU CMi's outer eccentricity differs by 2.6σ when SWINGERsim includes the classical apsidal advance; with point masses the fit agrees, and the published apsidal periods match SWINGERsim's point-mass integration to 1 %. The paper estimates a tidal advance of the same size but does not say which tidal term its integration used, so the cause remains open; freeing the apsidal constants does not remove it (the minima then want $k_2 \leq 0.0023$ for pair A against 0.0042 of the models). TIC 120362137's published apsidal period is a theoretical value. V994 Her's classical apsidal rates were fitted freely in the paper and differ from stellar structure; for pair B the authors found the same themselves (k_2 about 0.83 times the theoretical value), the large difference is pair C, which a companion of 0.13–0.31 $M_\odot$ on an 80–182-d orbit could explain.
6. **Data that need a second look:** the radial velocities of BU CMi (17–25 km/s scatter), the A–B eccentricity of V994 Her (0.51–0.59 from the minima against 0.69), and the depths of TESS sector 26 of V994 Her in the SPOC photometry (probably the crowding correction).

9 Data and reproducibility

The configurations are in `configs/squadra_test/` (published, anchored, fit and ensemble files for each system), the measured minima, radial velocities and TESS light curves in `data/observed/<system>/`, and the outputs of the fits in `output/squadra_test/<system>/`. Figures and numbers are made by `python tools/squadra_report.py` and `python tools/squadra_fits.py`; the oscillation of V994 Her is removed by `tools/remove_oscillation.py`, the SPOC crowding values are read by `tools/tess_crowding.py`. Parts I, III and IV of the test series are separate documents (*Testing SWINGERsim*, Parts I, III, IV).

References

- Borkovits T., Rappaport S., Kaye T., Isaacson H., Vanderburg A., Howard A. W., et al. (2019). Photodynamical analysis of the triply eclipsing hierarchical triple system EPIC 249432662. *MNRAS* 483, 1934. arXiv:1809.04366
- Borkovits T., Rappaport S. A., Hajdu T., Maxted P. F. L., Pál A., Forgács-Dajka E., Klagyivik P., Mitnyan T. (2020). TICs 167692429 and 220397947: the first compact hierarchical triple stars discovered with TESS. *MNRAS* 493, 5005.
- Borkovits T., et al. (2021). *MNRAS* 503, 3759.
- Borkovits T., Rappaport S. A., Chen H.-L., Torres G., Mitnyan T., Kostov V. B., Powell B. P., Pribulla T., Zasche P., et al. (2026). Discovery of the most compact 3+1-type quadruple star system TIC 120362137. *Nat. Commun.* 17, 1859. doi:10.1038/s41467-026-69223-4
- Choi J., Dotter A., Conroy C., Cantiello M., Paxton B., Johnson B. D. (2016). Mesa Isochrones and Stellar Tracks (MIST). I. Solar-scaled models. *ApJ* 823, 102. doi:10.3847/0004-637X/823/2/102
- Claret A. (2000). A new non-linear limb-darkening law for LTE stellar atmosphere models. *A&A* 363, 1081.
- Claret A. (2023). Theoretical tidal evolution constants for stellar models from the pre-main sequence to the white dwarf stage. *A&A* 674, A67. doi:10.1051/0004-6361/202346250
- Claret A., Bloemen S. (2011). Gravity and limb-darkening coefficients for the Kepler, CoRoT, Spitzer, uvby, UBVRIJHK, and Sloan photometric systems. *A&A* 529, A75. doi:10.1051/0004-6361/201116451
- Dotter A. (2016). MESA Isochrones and Stellar Tracks (MIST) 0: methods for the construction of stellar isochrones. *ApJS* 222, 8. doi:10.3847/0067-0049/222/1/8
- emcee documentation. Autocorrelation analysis & convergence (tutorial). emcee.readthedocs.io/en/s accessed 2026 November.
- Foreman-Mackey D., Hogg D. W., Lang D., Goodman J. (2013). emcee: the MCMC hammer. *PASP* 125, 306. doi:10.1086/670067
- Huang C. X., Vanderburg A., Pál A., et al. (2020). Photometry of 10 million stars from

the first two years of TESS full frame images: Part I. *RNAAS* 4, 204. doi:10.3847/2515-5172/abca2e
- Irwin J. B. (1952). The determination of a light-time orbit. *ApJ* 116, 211. doi:10.1086/145604
- Jenkins J. M., Twicken J. D., McCauliff S., et al. (2016). The TESS science processing operations center. *Proc. SPIE* 9913, 99133E. doi:10.1117/12.2233418
- Lee C.-U., Kim S.-L., Lee J. W., Kim C.-H., Jeon Y.-B., Kim H.-I., Yoon J.-N., Humphrey A. (2008). V994 Herculis: the multiple system with a quadruple-lined spectrum and a double eclipsing feature. *MNRAS* 389, 1630. doi:10.1111/j.1365-2966.2008.13708.x
- Lightkurve Collaboration (2018). Lightkurve: Kepler and TESS time series analysis in Python. Astrophysics Source Code Library, ascl:1812.013.
- Mardling R. A., Aarseth S. J. (2001). Tidal interactions in star cluster simulations. *MNRAS* 321, 398. doi:10.1046/j.1365-8711.2001.03974.x
- Ogloza W., Niewiadomski W., Barnacka A., Biskup M., Malek K., Sokolowski M. (2008). *Inf. Bull. Var. Stars* 5843, 1.
- Pont F., Zucker S., Queloz D. (2006). The effect of red noise on planetary transit detection. *MNRAS* 373, 231. doi:10.1111/j.1365-2966.2006.11012.x
- Pribulla T., Borkovits T., Jayaraman R., Rappaport S., Mitnyan T., Zasche P., Komžík R., Pál A., Uhlař R., Mašek M., Henzl Z., et al. (2023). BU Canis Minoris – the most compact known flat doubly eclipsing quadruple system. *MNRAS* 524, 4220. doi:10.1093/mnras/stad2015
- Rein H., Liu S.-F. (2012). REBOUND: an open-source multi-purpose N-body code for collisional dynamics. *A&A* 537, A128. doi:10.1051/0004-6361/201118085
- Rein H., Spiegel D. S. (2015). IAS15: a fast, adaptive, high-order integrator for gravitational dynamics. *MNRAS* 446, 1424. doi:10.1093/mnras/stu2164
- Ricker G. R., et al. (2015). Transiting Exoplanet Survey Satellite. *JATIS* 1, 014003. doi:10.1117/1.JATIS.1.1.014003
- Tamayo D., Rein H., Shi P., Hernandez D. M. (2020). REBOUNDx: a library for adding conservative and dissipative forces to otherwise symplectic N-body integrations. *MNRAS* 491, 2885. doi:10.1093/mnras/stz2870
- Virtanen P., et al. (2020). SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nat. Methods* 17, 261. doi:10.1038/s41592-019-0686-2
- Volkov I. M., Kravtsova A. S., Chochol D. (2021). *Astron. Rep.* 65, 826.
- Vynatheya P., Hamers A. S., Mardling R. A., Bellinger E. P. (2022). Algebraic and machine learning approach to hierarchical triple-star stability. *MNRAS* 516, 4146. arXiv:2207.03151
- Vynatheya P., Mardling R. A., Hamers A. S. (2023). Quadruple-star systems are not always nested triples: a machine learning approach to dynamical stability. *MNRAS* 525, 2388. arXiv:2301.09930
- Zasche P., Borkovits T., Jayaraman R., Rappaport S. A., Brož M., Vokrouhlický D., Bíró I. B., Hegedüs T., Kiss Z. T., Uhlař R., et al. (2023). V994 Herculis: a unique triply eclipsing sextuple star system. *MNRAS* 520, 3127. doi:10.1093/mnras/stad328